

Sealed Experiment Record

Sealed 2026-10-05 · Signed by the K-Veritas server
Session ca7dc592 · Data hash 3d35...f9a6

Abstract

This record seals 4 runs (kveritas_arm.py, apo.yaml, 0.1, apo.yaml) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-05 07:29 and 08:35 UTC. Reported results: phi 0.849638, mer 5.16667, mcs 0.865169, uwec 0.769663. 4 invocations in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	ca7dc592-2ee6-408e-812c-242702f7472a
Initialized	2026-10-05T07:29:03Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1
Environment, run 2	7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1
Environment, run 3	7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1
Environment, run 4	7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1
Disclosure	open
Source code, run 1	5785562323e0b346a317ca664df01d4f79267bf1b56030c3ef22d4cc0d5d5492
Source code, run 2	5785562323e0b346a317ca664df01d4f79267bf1b56030c3ef22d4cc0d5d5492
Source code, run 3	5785562323e0b346a317ca664df01d4f79267bf1b56030c3ef22d4cc0d5d5492
Source code, run 4	5785562323e0b346a317ca664df01d4f79267bf1b56030c3ef22d4cc0d5d5492

2 Runs

#	Command	Duration	Exit	Anchored
1	python cookbooks/kveritas_arm.py apo /tmp/k...	1.8 seconds	0	at run end
2	sebeni distill -c cookbooks/kveritas/apo.yaml	51.8 seconds	0	at run end
3	sebeni train -c cookbooks/kveritas/apo.yaml...	1 hours 3 minutes	0	at run end
4	sebeni eval -c cookbooks/kveritas/apo.yaml	2 minutes 3 seconds	0	at run end

4 invocations in total: 4 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kveritas_arm.py

No metrics reported.

Run 2: apo.yaml

No metrics reported.

Run 3: 0.1

Metric	Final value	Step	Source
loss	0.899064	500 (500 readings)	explicit
rewards_chosen	0.491311	500 (500 readings)	explicit
rewards_rejected	0.0567837	500 (500 readings)	explicit
rewards_accuracies	0.8125	500 (500 readings)	explicit
rewards_margins	0.434527	500 (500 readings)	explicit
eval_rewards_chosen	0.295494	500 (20 readings)	explicit
eval_rewards_rejected	-0.0857438	500 (20 readings)	explicit
eval_rewards_accuracies	0.669444	500 (20 readings)	explicit
eval_rewards_margins	0.381238	500 (20 readings)	explicit
eval_loss	0.911244	500 (10 readings)	explicit
eval_runtime	77.3757	500 (10 readings)	explicit
eval_samples_per_second	4.653	500 (10 readings)	explicit
eval_steps_per_second	0.582	500 (10 readings)	explicit
eval_entropy	0.887046	500 (10 readings)	explicit
eval_num_tokens	3.0435e+07	500 (10 readings)	explicit
eval_logits_chosen	10.1392	500 (10 readings)	explicit
eval_logits_rejected	10.1657	500 (10 readings)	explicit
eval_mean_token_accuracy	0.85068	500 (10 readings)	explicit
eval_logps_chosen	-1109.39	500 (10 readings)	explicit
eval_logps_rejected	-1111.44	500 (10 readings)	explicit
epoch	2.4642	500 (10 readings)	explicit
phi	0.847792	500 (10 readings)	explicit
phi_bam	0.947662	500 (10 readings)	explicit
phi_bmq	0.788389	500 (10 readings)	explicit
phi_mku	0.874554	500 (10 readings)	explicit
phi_myk	0.858518	500 (10 readings)	explicit
phi_ses	0.673515	500 (10 readings)	explicit
phi_dtm	0.875799	500 (10 readings)	explicit
phi_snk	0.839542	500 (10 readings)	explicit
phi_boz	0.880553	500 (10 readings)	explicit
phi_ful	0.872129	500 (10 readings)	explicit
phi_spp	0.999123	500 (10 readings)	explicit
phi_taq	0.715931	500 (10 readings)	explicit

Seed committed: seed:42, output line 1, 2026-10-05T07:29:59Z.

Run 4: apo.yaml

Metric	Final value	Step	Source
phi_bam	0.823658	0	explicit
mer_bam	0	0	explicit
mcs_bam	0	0	explicit

Metric	Final value	Step	Source
uwec_bam	1	0	explicit
phi_bmq	0.752429	0	explicit
mer_bmq	3	0	explicit
mcs_bmq	0.833333	0	explicit
uwec_bmq	0.666667	0	explicit
phi_boz	0.916823	0	explicit
mer_boz	0	0	explicit
mcs_boz	0	0	explicit
uwec_boz	1	0	explicit
phi_dtm	0.809387	0	explicit
mer_dtm	0.5	0	explicit
mcs_dtm	0.333333	0	explicit
uwec_dtm	0.666667	0	explicit
phi_ful	0.980054	0	explicit
mer_ful	3	0	explicit
mcs_ful	0.75	0	explicit
uwec_ful	0.625	0	explicit
phi_mey	0.768726	0	explicit
mer_mey	0	0	explicit
mcs_mey	0	0	explicit
uwec_mey	0.5	0	explicit
phi_mku	0.908601	0	explicit
mer_mku	12	0	explicit
mcs_mku	0.923077	0	explicit
uwec_mku	0.538462	0	explicit
phi_kao	0.970238	0	explicit
mer_kao	25.3333	0	explicit
mcs_kao	0.974359	0	explicit
uwec_kao	1	0	explicit
phi_myk	0.7953	0	explicit
mer_myk	10	0	explicit
mcs_myk	0.909091	0	explicit
uwec_myk	0.5	0	explicit
phi_ses	0.836952	0	explicit
mer_ses	0	0	explicit
mcs_ses	0	0	explicit
uwec_ses	1	0	explicit
phi_spp	0.853018	0	explicit
mer_spp	6.66667	0	explicit
mcs_spp	0.909091	0	explicit
uwec_spp	0.545455	0	explicit
phi_snk	0.781673	0	explicit
mer_snk	0.333333	0	explicit
mcs_snk	0.333333	0	explicit
uwec_snk	0.666667	0	explicit
phi_taq	0.84844	0	explicit
mer_taq	0.5	0	explicit
mcs_taq	0.5	0	explicit
uwec_taq	1	0	explicit
phi	0.849638	0	explicit

Metric	Final value	Step	Source
mer	5.16667	0	explicit
mcs	0.865169	0	explicit
uwec	0.769663	0	explicit

Inline claim	Value	Phase	Output line
phi	0.849638	eval	64
mer	5.16667	eval	66
mcs	0.865169	eval	68
uwec	0.769663	eval	70

4 Execution evidence

Coherence

Coherent (0.31): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	16	2.5 s	0.73 GB	0%
2	510	52.6 s	1.38 GB	0%
3	7200	3826.1 s	4.00 GB	99%
4	1229	123.9 s	3.38 GB	10%

5 Provenance

Run 1: kveritas_arm.py

Disclosure open. 137 tracked files, 2 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	07:29:03	1f46ebf2f29b9b43	136
1	run_end	07:29:04	741ab980f318aa55	1

Activity: 136 files read, 1 files written, 1 subprocesses. Full list in Appendix B.

Run 2: apo.yaml

Disclosure open. 137 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	07:29:05	741ab980f318aa55	137
2	run_end	07:29:57	741ab980f318aa55	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 160 files read, 0 files written, 0 subprocesses. Full list in Appendix B.

Run 3: 0.1

Disclosure open. 137 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	07:29:58	741ab980f318aa55	137
2	run_end	08:33:35	741ab980f318aa55	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 139 files read, 0 files written, 0 subprocesses. Full list in Appendix B.

Run 4: apo.yaml

Disclosure open. 137 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	08:33:36	741ab980f318aa55	137
2	run_end	08:35:39	741ab980f318aa55	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 141 files read, 0 files written, 0 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-05T08:35:40.609502+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	3d35b38ed6fb3a13c0ac03d020b070894bb4f1d5469f322ef456aca5b4eff9a6
Code bundle hash	62f040330bb576322043e72745134b995ff5b696bdfa1f8c62fa006027af6947

Run anchors: 4 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kveritas_arm.py

Run 2: apo.yaml

Phase	Line	Time	CPU time	Memory	GPU
distill	1	07:29:07	2.4 s	0.74 GB	

Run 3: 0.1

Metric	Value	Step	Line	Source
loss	1	1	13	explicit
rewards_chosen	0	1	14	explicit
rewards_rejected	0	1	15	explicit
rewards accuracies	0	1	16	explicit
rewards margins	0	1	17	explicit
loss	0.990924	2	19	explicit
rewards_chosen	-0.0290752	2	20	explicit
rewards_rejected	-0.0672332	2	21	explicit
rewards accuracies	0.375	2	22	explicit
rewards margins	0.0381579	2	23	explicit
loss	0.978554	3	25	explicit
rewards_chosen	-0.0343121	3	26	explicit
rewards_rejected	-0.121013	3	27	explicit
rewards accuracies	0.8125	3	28	explicit
rewards margins	0.0867014	3	29	explicit
loss	1.01712	4	31	explicit
rewards_chosen	-0.0188786	4	32	explicit
rewards_rejected	0.050581	4	33	explicit
rewards accuracies	0.3125	4	34	explicit
rewards margins	-0.0694595	4	35	explicit
loss	1.00752	5	37	explicit
rewards_chosen	-0.0777143	5	38	explicit
rewards_rejected	-0.0469396	5	39	explicit
rewards accuracies	0.6875	5	40	explicit
rewards margins	-0.0307747	5	41	explicit
loss	0.995448	6	43	explicit
rewards_chosen	-0.0381964	6	44	explicit
rewards_rejected	-0.055994	6	45	explicit
rewards accuracies	0.5	6	46	explicit
rewards margins	0.0177977	6	47	explicit
loss	0.992572	7	49	explicit
rewards_chosen	-0.0512753	7	50	explicit
rewards_rejected	-0.0812232	7	51	explicit
rewards accuracies	0.5	7	52	explicit
rewards margins	0.0299479	7	53	explicit
loss	0.99978	8	55	explicit
rewards_chosen	-0.124762	8	56	explicit
rewards_rejected	-0.12653	8	57	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.4375	8	58	explicit
rewards_margins	0.00176754	8	59	explicit
loss	1.02031	9	61	explicit
rewards_chosen	-0.0564267	9	62	explicit
rewards_rejected	0.025969	9	63	explicit
rewards_accuracies	0.5	9	64	explicit
rewards_margins	-0.0823957	9	65	explicit
loss	1.01638	10	67	explicit
rewards_chosen	-0.00183754	10	68	explicit
rewards_rejected	0.0642998	10	69	explicit
rewards_accuracies	0.375	10	70	explicit
rewards_margins	-0.0661373	10	71	explicit
loss	0.991386	11	73	explicit
rewards_chosen	-0.0924952	11	74	explicit
rewards_rejected	-0.127224	11	75	explicit
rewards_accuracies	0.625	11	76	explicit
rewards_margins	0.0347292	11	77	explicit
loss	1.01676	12	79	explicit
rewards_chosen	-0.0652061	12	80	explicit
rewards_rejected	0.00321369	12	81	explicit
rewards_accuracies	0.5	12	82	explicit
rewards_margins	-0.0684198	12	83	explicit
loss	0.969625	13	85	explicit
rewards_chosen	0.0381334	13	86	explicit
rewards_rejected	-0.0849941	13	87	explicit
rewards_accuracies	0.6875	13	88	explicit
rewards_margins	0.123128	13	89	explicit
loss	1.01547	14	91	explicit
rewards_chosen	-0.0851307	14	92	explicit
rewards_rejected	-0.0226971	14	93	explicit
rewards_accuracies	0.3125	14	94	explicit
rewards_margins	-0.0624336	14	95	explicit
loss	0.995832	15	97	explicit
rewards_chosen	0.0646378	15	98	explicit
rewards_rejected	0.0475426	15	99	explicit
rewards_accuracies	0.375	15	100	explicit
rewards_margins	0.0170952	15	101	explicit
loss	1.01797	16	103	explicit
rewards_chosen	-0.101042	16	104	explicit
rewards_rejected	-0.0279514	16	105	explicit
rewards_accuracies	0.3125	16	106	explicit
rewards_margins	-0.0730907	16	107	explicit
loss	0.966155	17	109	explicit
rewards_chosen	-0.0504227	17	110	explicit
rewards_rejected	-0.187832	17	111	explicit
rewards_accuracies	0.75	17	112	explicit
rewards_margins	0.13741	17	113	explicit
loss	1.0124	18	115	explicit
rewards_chosen	-0.0309444	18	116	explicit
rewards_rejected	0.0188251	18	117	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.4375	18	118	explicit
rewards_margins	-0.0497696	18	119	explicit
loss	0.978303	19	121	explicit
rewards_chosen	0.0325697	19	122	explicit
rewards_rejected	-0.0550821	19	123	explicit
rewards_accuracies	0.625	19	124	explicit
rewards_margins	0.0876518	19	125	explicit
loss	1.00181	20	127	explicit
rewards_chosen	0.0544476	20	128	explicit
rewards_rejected	0.0622772	20	129	explicit
rewards_accuracies	0.3125	20	130	explicit
rewards_margins	-0.00782967	20	131	explicit
loss	0.977989	21	133	explicit
rewards_chosen	0.0546215	21	134	explicit
rewards_rejected	-0.0343937	21	135	explicit
rewards_accuracies	0.5625	21	136	explicit
rewards_margins	0.0890152	21	137	explicit
loss	1.00285	22	139	explicit
rewards_chosen	-0.0055502	22	140	explicit
rewards_rejected	0.00623951	22	141	explicit
rewards_accuracies	0.375	22	142	explicit
rewards_margins	-0.0117897	22	143	explicit
loss	0.981193	23	145	explicit
rewards_chosen	0.0934921	23	146	explicit
rewards_rejected	0.0185343	23	147	explicit
rewards_accuracies	0.6875	23	148	explicit
rewards_margins	0.0749578	23	149	explicit
loss	0.986334	24	151	explicit
rewards_chosen	0.0885334	24	152	explicit
rewards_rejected	0.0328541	24	153	explicit
rewards_accuracies	0.625	24	154	explicit
rewards_margins	0.0556793	24	155	explicit
loss	1.01433	25	157	explicit
rewards_chosen	-0.0389156	25	158	explicit
rewards_rejected	0.0194595	25	159	explicit
rewards_accuracies	0.4375	25	160	explicit
rewards_margins	-0.0583752	25	161	explicit
loss	0.9982	26	163	explicit
rewards_chosen	-0.0417763	26	164	explicit
rewards_rejected	-0.0490189	26	165	explicit
rewards_accuracies	0.4375	26	166	explicit
rewards_margins	0.00724258	26	167	explicit
loss	0.99983	27	169	explicit
rewards_chosen	0.063076	27	170	explicit
rewards_rejected	0.0624489	27	171	explicit
rewards_accuracies	0.375	27	172	explicit
rewards_margins	0.000627134	27	173	explicit
loss	1.00297	28	175	explicit
rewards_chosen	0.0665062	28	176	explicit
rewards_rejected	0.077721	28	177	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.1875	28	178	explicit
rewards_margins	-0.0112148	28	179	explicit
loss	0.974562	29	181	explicit
rewards_chosen	0.0871056	29	182	explicit
rewards_rejected	-0.0160046	29	183	explicit
rewards_accuracies	0.625	29	184	explicit
rewards_margins	0.10311	29	185	explicit
loss	0.99891	30	187	explicit
rewards_chosen	0.0523031	30	188	explicit
rewards_rejected	0.0480556	30	189	explicit
rewards_accuracies	0.375	30	190	explicit
rewards_margins	0.00424757	30	191	explicit
loss	0.983467	31	193	explicit
rewards_chosen	0.0814098	31	194	explicit
rewards_rejected	0.0127346	31	195	explicit
rewards_accuracies	0.5	31	196	explicit
rewards_margins	0.0686752	31	197	explicit
loss	1.0095	32	199	explicit
rewards_chosen	-0.0560322	32	200	explicit
rewards_rejected	-0.0183826	32	201	explicit
rewards_accuracies	0.4375	32	202	explicit
rewards_margins	-0.0376495	32	203	explicit
loss	0.987226	33	205	explicit
rewards_chosen	0.0627613	33	206	explicit
rewards_rejected	0.0111614	33	207	explicit
rewards_accuracies	0.4375	33	208	explicit
rewards_margins	0.0515999	33	209	explicit
loss	1.00094	34	211	explicit
rewards_chosen	0.0375343	34	212	explicit
rewards_rejected	0.0408063	34	213	explicit
rewards_accuracies	0.4375	34	214	explicit
rewards_margins	-0.00327196	34	215	explicit
loss	1.0188	35	217	explicit
rewards_chosen	0.0663107	35	218	explicit
rewards_rejected	0.142053	35	219	explicit
rewards_accuracies	0.375	35	220	explicit
rewards_margins	-0.0757423	35	221	explicit
loss	1.00562	36	223	explicit
rewards_chosen	0.121121	36	224	explicit
rewards_rejected	0.143776	36	225	explicit
rewards_accuracies	0.4375	36	226	explicit
rewards_margins	-0.0226547	36	227	explicit
loss	0.997035	37	229	explicit
rewards_chosen	0.0494727	37	230	explicit
rewards_rejected	0.0373956	37	231	explicit
rewards_accuracies	0.4375	37	232	explicit
rewards_margins	0.0120771	37	233	explicit
loss	0.99583	38	235	explicit
rewards_chosen	0.0389187	38	236	explicit
rewards_rejected	0.0219593	38	237	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.4375	38	238	explicit
rewards_margins	0.0169594	38	239	explicit
loss	0.997474	39	241	explicit
rewards_chosen	0.123672	39	242	explicit
rewards_rejected	0.112174	39	243	explicit
rewards_accuracies	0.5	39	244	explicit
rewards_margins	0.0114985	39	245	explicit
loss	1.01339	40	247	explicit
rewards_chosen	0.133998	40	248	explicit
rewards_rejected	0.187186	40	249	explicit
rewards_accuracies	0.375	40	250	explicit
rewards_margins	-0.0531876	40	251	explicit
loss	0.990064	41	253	explicit
rewards_chosen	0.065567	41	254	explicit
rewards_rejected	0.0262127	41	255	explicit
rewards_accuracies	0.5625	41	256	explicit
rewards_margins	0.0393543	41	257	explicit
loss	1.01374	42	259	explicit
rewards_chosen	0.111817	42	260	explicit
rewards_rejected	0.167065	42	261	explicit
rewards_accuracies	0.3125	42	262	explicit
rewards_margins	-0.0552479	42	263	explicit
loss	1.00611	43	265	explicit
rewards_chosen	0.211837	43	266	explicit
rewards_rejected	0.236736	43	267	explicit
rewards_accuracies	0.4375	43	268	explicit
rewards_margins	-0.0248997	43	269	explicit
loss	1.00097	44	271	explicit
rewards_chosen	0.0791683	44	272	explicit
rewards_rejected	0.0832016	44	273	explicit
rewards_accuracies	0.375	44	274	explicit
rewards_margins	-0.00403328	44	275	explicit
loss	0.983609	45	277	explicit
rewards_chosen	0.125959	45	278	explicit
rewards_rejected	0.059606	45	279	explicit
rewards_accuracies	0.5	45	280	explicit
rewards_margins	0.0663528	45	281	explicit
loss	1.00354	46	283	explicit
rewards_chosen	0.104315	46	284	explicit
rewards_rejected	0.118516	46	285	explicit
rewards_accuracies	0.375	46	286	explicit
rewards_margins	-0.014201	46	287	explicit
loss	0.988344	47	289	explicit
rewards_chosen	0.0148441	47	290	explicit
rewards_rejected	-0.0320117	47	291	explicit
rewards_accuracies	0.5625	47	292	explicit
rewards_margins	0.0468558	47	293	explicit
loss	0.989783	48	295	explicit
rewards_chosen	0.10284	48	296	explicit
rewards_rejected	0.0617142	48	297	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.5625	48	298	explicit
rewards_margins	0.0411255	48	299	explicit
loss	1.00234	49	301	explicit
rewards_chosen	0.037343	49	302	explicit
rewards_rejected	0.0471762	49	303	explicit
rewards_accuracies	0.4375	49	304	explicit
rewards_margins	-0.00983315	49	305	explicit
loss	0.992141	50	307	explicit
rewards_chosen	0.259343	50	308	explicit
rewards_rejected	0.227317	50	309	explicit
rewards_accuracies	0.4375	50	310	explicit
rewards_margins	0.0320255	50	311	explicit
eval_rewards_chosen	0.150473	50	313	explicit
eval_rewards_rejected	0.112928	50	314	explicit
eval_rewards_accuracies	0.494444	50	315	explicit
eval_rewards_margins	0.0375453	50	316	explicit
eval_loss	0.990799	50	318	explicit
eval_runtime	77.1861	50	319	explicit
eval_samples_per_second	4.664	50	320	explicit
eval_steps_per_second	0.583	50	321	explicit
eval_entropy	0.891719	50	322	explicit
eval_num_tokens	3.02391e+06	50	323	explicit
eval_logits_chosen	10.3631	50	324	explicit
eval_logits_rejected	10.3781	50	325	explicit
eval_mean_token_accuracy	0.850478	50	326	explicit
eval_rewards_chosen	0.150473	50	327	explicit
eval_rewards_rejected	0.112928	50	328	explicit
eval_rewards_accuracies	0.494444	50	329	explicit
eval_rewards_margins	0.0375453	50	330	explicit
eval_logps_chosen	-1110.84	50	331	explicit
eval_logps_rejected	-1109.46	50	332	explicit
epoch	0.246914	50	333	explicit
phi	0.847792	50	334	explicit
phi_bam	0.947662	50	335	explicit
phi_bmq	0.788389	50	336	explicit
phi_mku	0.874554	50	337	explicit
phi_myk	0.858518	50	338	explicit
phi_ses	0.673515	50	339	explicit
phi_dtm	0.875799	50	340	explicit
phi_snk	0.839542	50	341	explicit
phi_boz	0.880553	50	342	explicit
phi_ful	0.872129	50	343	explicit
phi_spp	0.999123	50	344	explicit
phi_taq	0.715931	50	345	explicit
loss	0.993833	51	346	explicit
rewards_chosen	0.140717	51	347	explicit
rewards_rejected	0.115698	51	348	explicit
rewards_accuracies	0.4375	51	349	explicit
rewards_margins	0.0250191	51	350	explicit
loss	0.996644	52	352	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.137482	52	353	explicit
rewards_rejected	0.124865	52	354	explicit
rewards_accuracies	0.375	52	355	explicit
rewards_margins	0.0126167	52	356	explicit
loss	0.980858	53	358	explicit
rewards_chosen	0.145601	53	359	explicit
rewards_rejected	0.068504	53	360	explicit
rewards_accuracies	0.75	53	361	explicit
rewards_margins	0.0770966	53	362	explicit
loss	0.97607	54	364	explicit
rewards_chosen	0.129043	54	365	explicit
rewards_rejected	0.0323677	54	366	explicit
rewards_accuracies	0.5625	54	367	explicit
rewards_margins	0.0966755	54	368	explicit
loss	0.980684	55	370	explicit
rewards_chosen	0.092931	55	371	explicit
rewards_rejected	0.0146579	55	372	explicit
rewards_accuracies	0.5625	55	373	explicit
rewards_margins	0.0782731	55	374	explicit
loss	0.966073	56	376	explicit
rewards_chosen	0.266927	56	377	explicit
rewards_rejected	0.129342	56	378	explicit
rewards_accuracies	0.5625	56	379	explicit
rewards_margins	0.137585	56	380	explicit
loss	0.983141	57	382	explicit
rewards_chosen	0.178876	57	383	explicit
rewards_rejected	0.110105	57	384	explicit
rewards_accuracies	0.5625	57	385	explicit
rewards_margins	0.0687718	57	386	explicit
loss	0.983913	58	388	explicit
rewards_chosen	0.138531	58	389	explicit
rewards_rejected	0.0737147	58	390	explicit
rewards_accuracies	0.625	58	391	explicit
rewards_margins	0.0648167	58	392	explicit
loss	0.996797	59	394	explicit
rewards_chosen	0.266068	59	395	explicit
rewards_rejected	0.253174	59	396	explicit
rewards_accuracies	0.375	59	397	explicit
rewards_margins	0.0128944	59	398	explicit
loss	1.00042	60	400	explicit
rewards_chosen	0.138313	60	401	explicit
rewards_rejected	0.140672	60	402	explicit
rewards_accuracies	0.4375	60	403	explicit
rewards_margins	-0.00235825	60	404	explicit
loss	0.984788	61	406	explicit
rewards_chosen	0.153771	61	407	explicit
rewards_rejected	0.0926594	61	408	explicit
rewards_accuracies	0.625	61	409	explicit
rewards_margins	0.061112	61	410	explicit
loss	0.962823	62	412	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.248872	62	413	explicit
rewards_rejected	0.09776	62	414	explicit
rewards_accuracies	0.75	62	415	explicit
rewards_margins	0.151112	62	416	explicit
loss	0.97741	63	418	explicit
rewards_chosen	0.151141	63	419	explicit
rewards_rejected	0.0601675	63	420	explicit
rewards_accuracies	0.625	63	421	explicit
rewards_margins	0.0909731	63	422	explicit
loss	0.978082	64	424	explicit
rewards_chosen	0.152545	64	425	explicit
rewards_rejected	0.0626778	64	426	explicit
rewards_accuracies	0.5625	64	427	explicit
rewards_margins	0.0898674	64	428	explicit
loss	0.998313	65	430	explicit
rewards_chosen	0.117289	65	431	explicit
rewards_rejected	0.109586	65	432	explicit
rewards_accuracies	0.625	65	433	explicit
rewards_margins	0.00770359	65	434	explicit
loss	0.97405	66	436	explicit
rewards_chosen	0.123056	66	437	explicit
rewards_rejected	0.0179039	66	438	explicit
rewards_accuracies	0.625	66	439	explicit
rewards_margins	0.105152	66	440	explicit
loss	0.971432	67	442	explicit
rewards_chosen	0.345411	67	443	explicit
rewards_rejected	0.228121	67	444	explicit
rewards_accuracies	0.6875	67	445	explicit
rewards_margins	0.11729	67	446	explicit
loss	0.983464	68	448	explicit
rewards_chosen	0.208142	68	449	explicit
rewards_rejected	0.138887	68	450	explicit
rewards_accuracies	0.5625	68	451	explicit
rewards_margins	0.0692558	68	452	explicit
loss	0.982349	69	454	explicit
rewards_chosen	0.129411	69	455	explicit
rewards_rejected	0.0567339	69	456	explicit
rewards_accuracies	0.5	69	457	explicit
rewards_margins	0.0726767	69	458	explicit
loss	0.982393	70	460	explicit
rewards_chosen	0.214089	70	461	explicit
rewards_rejected	0.141921	70	462	explicit
rewards_accuracies	0.625	70	463	explicit
rewards_margins	0.0721678	70	464	explicit
loss	0.983761	71	466	explicit
rewards_chosen	0.168216	71	467	explicit
rewards_rejected	0.102213	71	468	explicit
rewards_accuracies	0.5625	71	469	explicit
rewards_margins	0.0660027	71	470	explicit
loss	0.969274	72	472	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.205113	72	473	explicit
rewards_rejected	0.0808178	72	474	explicit
rewards_accuracies	0.5625	72	475	explicit
rewards_margins	0.124295	72	476	explicit
loss	0.998289	73	478	explicit
rewards_chosen	0.168483	73	479	explicit
rewards_rejected	0.161463	73	480	explicit
rewards_accuracies	0.5625	73	481	explicit
rewards_margins	0.00702056	73	482	explicit
loss	0.986869	74	484	explicit
rewards_chosen	0.17128	74	485	explicit
rewards_rejected	0.11751	74	486	explicit
rewards_accuracies	0.625	74	487	explicit
rewards_margins	0.0537695	74	488	explicit
loss	0.998754	75	490	explicit
rewards_chosen	0.144519	75	491	explicit
rewards_rejected	0.139383	75	492	explicit
rewards_accuracies	0.375	75	493	explicit
rewards_margins	0.00513649	75	494	explicit
loss	0.98478	76	496	explicit
rewards_chosen	0.268545	76	497	explicit
rewards_rejected	0.20718	76	498	explicit
rewards_accuracies	0.5625	76	499	explicit
rewards_margins	0.0613657	76	500	explicit
loss	0.991178	77	502	explicit
rewards_chosen	0.198536	77	503	explicit
rewards_rejected	0.162856	77	504	explicit
rewards_accuracies	0.5625	77	505	explicit
rewards_margins	0.0356802	77	506	explicit
loss	0.990256	78	508	explicit
rewards_chosen	0.198957	78	509	explicit
rewards_rejected	0.159155	78	510	explicit
rewards_accuracies	0.5	78	511	explicit
rewards_margins	0.0398026	78	512	explicit
loss	0.987441	79	514	explicit
rewards_chosen	0.256536	79	515	explicit
rewards_rejected	0.206231	79	516	explicit
rewards_accuracies	0.5	79	517	explicit
rewards_margins	0.0503048	79	518	explicit
loss	0.992418	80	520	explicit
rewards_chosen	0.225163	80	521	explicit
rewards_rejected	0.195099	80	522	explicit
rewards_accuracies	0.4375	80	523	explicit
rewards_margins	0.0300632	80	524	explicit
loss	0.973085	81	526	explicit
rewards_chosen	0.168766	81	527	explicit
rewards_rejected	0.0604786	81	528	explicit
rewards_accuracies	0.625	81	529	explicit
rewards_margins	0.108287	81	530	explicit
loss	0.999474	82	532	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.0957223	82	533	explicit
rewards_rejected	0.0944106	82	534	explicit
rewards_accuracies	0.5625	82	535	explicit
rewards_margins	0.00131168	82	536	explicit
loss	0.985373	83	538	explicit
rewards_chosen	0.179663	83	539	explicit
rewards_rejected	0.119278	83	540	explicit
rewards_accuracies	0.375	83	541	explicit
rewards_margins	0.0603852	83	542	explicit
loss	0.959256	84	544	explicit
rewards_chosen	0.318781	84	545	explicit
rewards_rejected	0.152106	84	546	explicit
rewards_accuracies	0.8125	84	547	explicit
rewards_margins	0.166675	84	548	explicit
loss	0.95587	85	550	explicit
rewards_chosen	0.246139	85	551	explicit
rewards_rejected	0.0652609	85	552	explicit
rewards_accuracies	0.75	85	553	explicit
rewards_margins	0.180878	85	554	explicit
loss	0.96824	86	556	explicit
rewards_chosen	0.219268	86	557	explicit
rewards_rejected	0.0883621	86	558	explicit
rewards_accuracies	0.625	86	559	explicit
rewards_margins	0.130906	86	560	explicit
loss	0.973172	87	562	explicit
rewards_chosen	0.324028	87	563	explicit
rewards_rejected	0.21011	87	564	explicit
rewards_accuracies	0.5	87	565	explicit
rewards_margins	0.113918	87	566	explicit
loss	1.00015	88	568	explicit
rewards_chosen	0.126674	88	569	explicit
rewards_rejected	0.126759	88	570	explicit
rewards_accuracies	0.5625	88	571	explicit
rewards_margins	-8.4307e-05	88	572	explicit
loss	0.993563	89	574	explicit
rewards_chosen	0.215416	89	575	explicit
rewards_rejected	0.191674	89	576	explicit
rewards_accuracies	0.4375	89	577	explicit
rewards_margins	0.0237418	89	578	explicit
loss	0.966245	90	580	explicit
rewards_chosen	0.271275	90	581	explicit
rewards_rejected	0.131044	90	582	explicit
rewards_accuracies	0.6875	90	583	explicit
rewards_margins	0.140231	90	584	explicit
loss	0.977865	91	586	explicit
rewards_chosen	0.185297	91	587	explicit
rewards_rejected	0.0960686	91	588	explicit
rewards_accuracies	0.625	91	589	explicit
rewards_margins	0.0892288	91	590	explicit
loss	0.986567	92	592	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.201559	92	593	explicit
rewards_rejected	0.145817	92	594	explicit
rewards_accuracies	0.625	92	595	explicit
rewards_margins	0.0557421	92	596	explicit
loss	0.950583	93	598	explicit
rewards_chosen	0.237468	93	599	explicit
rewards_rejected	0.0366526	93	600	explicit
rewards_accuracies	0.8125	93	601	explicit
rewards_margins	0.200815	93	602	explicit
loss	0.95828	94	604	explicit
rewards_chosen	0.145472	94	605	explicit
rewards_rejected	-0.0232403	94	606	explicit
rewards_accuracies	0.875	94	607	explicit
rewards_margins	0.168712	94	608	explicit
loss	1.00253	95	610	explicit
rewards_chosen	0.270323	95	611	explicit
rewards_rejected	0.279534	95	612	explicit
rewards_accuracies	0.5	95	613	explicit
rewards_margins	-0.00921097	95	614	explicit
loss	0.981737	96	616	explicit
rewards_chosen	0.224503	96	617	explicit
rewards_rejected	0.151235	96	618	explicit
rewards_accuracies	0.5	96	619	explicit
rewards_margins	0.0732686	96	620	explicit
loss	0.983036	97	622	explicit
rewards_chosen	0.200813	97	623	explicit
rewards_rejected	0.133239	97	624	explicit
rewards_accuracies	0.5	97	625	explicit
rewards_margins	0.0675739	97	626	explicit
loss	0.970231	98	628	explicit
rewards_chosen	0.231214	98	629	explicit
rewards_rejected	0.109762	98	630	explicit
rewards_accuracies	0.6875	98	631	explicit
rewards_margins	0.121452	98	632	explicit
loss	0.956436	99	634	explicit
rewards_chosen	0.242614	99	635	explicit
rewards_rejected	0.0667332	99	636	explicit
rewards_accuracies	0.75	99	637	explicit
rewards_margins	0.175881	99	638	explicit
loss	0.978493	100	640	explicit
rewards_chosen	0.189553	100	641	explicit
rewards_rejected	0.101982	100	642	explicit
rewards_accuracies	0.4375	100	643	explicit
rewards_margins	0.0875713	100	644	explicit
eval_rewards_chosen	0.203812	100	646	explicit
eval_rewards_rejected	0.112071	100	647	explicit
eval_rewards_accuracies	0.583333	100	648	explicit
eval_rewards_margins	0.0917416	100	649	explicit
eval_loss	0.977382	100	651	explicit
eval_runtime	77.2097	100	652	explicit

Metric	Value	Step	Line	Source
eval_samples_per_second	4.663	100	653	explicit
eval_steps_per_second	0.583	100	654	explicit
eval_entropy	0.889787	100	655	explicit
eval_num_tokens	6.05624e+06	100	656	explicit
eval_logits_chosen	10.3271	100	657	explicit
eval_logits_rejected	10.3472	100	658	explicit
eval_mean_token_accuracy	0.850676	100	659	explicit
eval_rewards_chosen	0.203812	100	660	explicit
eval_rewards_rejected	0.112071	100	661	explicit
eval_rewards_accuracies	0.583333	100	662	explicit
eval_rewards_margins	0.0917416	100	663	explicit
eval_logps_chosen	-1110.31	100	664	explicit
eval_logps_rejected	-1109.46	100	665	explicit
epoch	0.493827	100	666	explicit
phi	0.847792	100	667	explicit
phi_bam	0.947662	100	668	explicit
phi_bmq	0.788389	100	669	explicit
phi_mku	0.874554	100	670	explicit
phi_myk	0.858518	100	671	explicit
phi_ses	0.673515	100	672	explicit
phi_dtm	0.875799	100	673	explicit
phi_snk	0.839542	100	674	explicit
phi_boz	0.880553	100	675	explicit
phi_ful	0.872129	100	676	explicit
phi_spp	0.999123	100	677	explicit
phi_taq	0.715931	100	678	explicit
loss	0.977696	101	679	explicit
rewards_chosen	0.173029	101	680	explicit
rewards_rejected	0.0833366	101	681	explicit
rewards_accuracies	0.5625	101	682	explicit
rewards_margins	0.0896927	101	683	explicit
loss	0.946754	102	685	explicit
rewards_chosen	0.21186	102	686	explicit
rewards_rejected	-0.00569196	102	687	explicit
rewards_accuracies	0.75	102	688	explicit
rewards_margins	0.217552	102	689	explicit
loss	0.946822	103	691	explicit
rewards_chosen	0.283748	103	692	explicit
rewards_rejected	0.0668797	103	693	explicit
rewards_accuracies	0.75	103	694	explicit
rewards_margins	0.216869	103	695	explicit
loss	0.954663	104	697	explicit
rewards_chosen	0.299868	104	698	explicit
rewards_rejected	0.114853	104	699	explicit
rewards_accuracies	0.8125	104	700	explicit
rewards_margins	0.185014	104	701	explicit
loss	0.981397	105	703	explicit
rewards_chosen	0.230856	105	704	explicit
rewards_rejected	0.155762	105	705	explicit
rewards_accuracies	0.5625	105	706	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.0750935	105	707	explicit
loss	0.970166	106	709	explicit
rewards_chosen	0.273049	106	710	explicit
rewards_rejected	0.1509	106	711	explicit
rewards_accuracies	0.5625	106	712	explicit
rewards_margins	0.122149	106	713	explicit
loss	0.934564	107	715	explicit
rewards_chosen	0.267904	107	716	explicit
rewards_rejected	-0.000471115	107	717	explicit
rewards_accuracies	0.75	107	718	explicit
rewards_margins	0.268375	107	719	explicit
loss	0.976094	108	721	explicit
rewards_chosen	0.173206	108	722	explicit
rewards_rejected	0.0754639	108	723	explicit
rewards_accuracies	0.5625	108	724	explicit
rewards_margins	0.0977421	108	725	explicit
loss	0.976082	109	727	explicit
rewards_chosen	0.245247	109	728	explicit
rewards_rejected	0.147987	109	729	explicit
rewards_accuracies	0.5625	109	730	explicit
rewards_margins	0.0972599	109	731	explicit
loss	0.955685	110	733	explicit
rewards_chosen	0.185091	110	734	explicit
rewards_rejected	0.00149497	110	735	explicit
rewards_accuracies	0.625	110	736	explicit
rewards_margins	0.183596	110	737	explicit
loss	0.97939	111	739	explicit
rewards_chosen	0.244999	111	740	explicit
rewards_rejected	0.161496	111	741	explicit
rewards_accuracies	0.5	111	742	explicit
rewards_margins	0.0835028	111	743	explicit
loss	0.990119	112	745	explicit
rewards_chosen	0.137947	112	746	explicit
rewards_rejected	0.0982361	112	747	explicit
rewards_accuracies	0.375	112	748	explicit
rewards_margins	0.0397104	112	749	explicit
loss	0.979137	113	751	explicit
rewards_chosen	0.222566	113	752	explicit
rewards_rejected	0.133391	113	753	explicit
rewards_accuracies	0.5625	113	754	explicit
rewards_margins	0.0891747	113	755	explicit
loss	0.970914	114	757	explicit
rewards_chosen	0.149149	114	758	explicit
rewards_rejected	0.0298788	114	759	explicit
rewards_accuracies	0.5	114	760	explicit
rewards_margins	0.11927	114	761	explicit
loss	0.962054	115	763	explicit
rewards_chosen	0.142892	115	764	explicit
rewards_rejected	-0.0128571	115	765	explicit
rewards_accuracies	0.5	115	766	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.155749	115	767	explicit
loss	0.996605	116	769	explicit
rewards_chosen	0.112345	116	770	explicit
rewards_rejected	0.0976177	116	771	explicit
rewards_accuracies	0.5	116	772	explicit
rewards_margins	0.0147274	116	773	explicit
loss	0.95036	117	775	explicit
rewards_chosen	0.253059	117	776	explicit
rewards_rejected	0.0484493	117	777	explicit
rewards_accuracies	0.75	117	778	explicit
rewards_margins	0.20461	117	779	explicit
loss	0.980424	118	781	explicit
rewards_chosen	0.199919	118	782	explicit
rewards_rejected	0.119947	118	783	explicit
rewards_accuracies	0.4375	118	784	explicit
rewards_margins	0.0799717	118	785	explicit
loss	0.991102	119	787	explicit
rewards_chosen	0.191203	119	788	explicit
rewards_rejected	0.154306	119	789	explicit
rewards_accuracies	0.6875	119	790	explicit
rewards_margins	0.036897	119	791	explicit
loss	0.965056	120	793	explicit
rewards_chosen	0.154779	120	794	explicit
rewards_rejected	0.0130788	120	795	explicit
rewards_accuracies	0.5	120	796	explicit
rewards_margins	0.1417	120	797	explicit
loss	0.966864	121	799	explicit
rewards_chosen	0.0822361	121	800	explicit
rewards_rejected	-0.0521982	121	801	explicit
rewards_accuracies	0.625	121	802	explicit
rewards_margins	0.134434	121	803	explicit
loss	0.92558	122	805	explicit
rewards_chosen	0.254853	122	806	explicit
rewards_rejected	-0.0480221	122	807	explicit
rewards_accuracies	0.8125	122	808	explicit
rewards_margins	0.302876	122	809	explicit
loss	0.963866	123	811	explicit
rewards_chosen	0.0671101	123	812	explicit
rewards_rejected	-0.0802483	123	813	explicit
rewards_accuracies	0.625	123	814	explicit
rewards_margins	0.147358	123	815	explicit
loss	0.959144	124	817	explicit
rewards_chosen	0.176774	124	818	explicit
rewards_rejected	0.00901795	124	819	explicit
rewards_accuracies	0.4375	124	820	explicit
rewards_margins	0.167756	124	821	explicit
loss	0.990769	125	823	explicit
rewards_chosen	0.114536	125	824	explicit
rewards_rejected	0.0788612	125	825	explicit
rewards_accuracies	0.4375	125	826	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.0356745	125	827	explicit
loss	0.972048	126	829	explicit
rewards_chosen	0.140196	126	830	explicit
rewards_rejected	0.0270767	126	831	explicit
rewards_accuracies	0.625	126	832	explicit
rewards_margins	0.11312	126	833	explicit
loss	0.941282	127	835	explicit
rewards_chosen	0.192737	127	836	explicit
rewards_rejected	-0.0479057	127	837	explicit
rewards_accuracies	0.75	127	838	explicit
rewards_margins	0.240643	127	839	explicit
loss	0.9179	128	841	explicit
rewards_chosen	0.218789	128	842	explicit
rewards_rejected	-0.114828	128	843	explicit
rewards_accuracies	0.8125	128	844	explicit
rewards_margins	0.333617	128	845	explicit
loss	0.967013	129	847	explicit
rewards_chosen	0.132586	129	848	explicit
rewards_rejected	-0.00261192	129	849	explicit
rewards_accuracies	0.75	129	850	explicit
rewards_margins	0.135198	129	851	explicit
loss	0.956108	130	853	explicit
rewards_chosen	0.158502	130	854	explicit
rewards_rejected	-0.0215389	130	855	explicit
rewards_accuracies	0.5625	130	856	explicit
rewards_margins	0.180041	130	857	explicit
loss	0.952291	131	859	explicit
rewards_chosen	0.0107449	131	860	explicit
rewards_rejected	-0.184369	131	861	explicit
rewards_accuracies	0.625	131	862	explicit
rewards_margins	0.195114	131	863	explicit
loss	0.947679	132	865	explicit
rewards_chosen	0.149267	132	866	explicit
rewards_rejected	-0.0631687	132	867	explicit
rewards_accuracies	0.5	132	868	explicit
rewards_margins	0.212436	132	869	explicit
loss	0.928243	133	871	explicit
rewards_chosen	0.148794	133	872	explicit
rewards_rejected	-0.144506	133	873	explicit
rewards_accuracies	0.875	133	874	explicit
rewards_margins	0.2933	133	875	explicit
loss	0.953607	134	877	explicit
rewards_chosen	0.293578	134	878	explicit
rewards_rejected	0.100995	134	879	explicit
rewards_accuracies	0.75	134	880	explicit
rewards_margins	0.192582	134	881	explicit
loss	0.965526	135	883	explicit
rewards_chosen	0.194917	135	884	explicit
rewards_rejected	0.0546795	135	885	explicit
rewards_accuracies	0.5625	135	886	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.140238	135	887	explicit
loss	0.973254	136	889	explicit
rewards_chosen	0.116346	136	890	explicit
rewards_rejected	0.00616837	136	891	explicit
rewards_accuracies	0.5625	136	892	explicit
rewards_margins	0.110178	136	893	explicit
loss	0.94228	137	895	explicit
rewards_chosen	0.0484739	137	896	explicit
rewards_rejected	-0.184748	137	897	explicit
rewards_accuracies	0.75	137	898	explicit
rewards_margins	0.233222	137	899	explicit
loss	0.98977	138	901	explicit
rewards_chosen	0.142551	138	902	explicit
rewards_rejected	0.099704	138	903	explicit
rewards_accuracies	0.625	138	904	explicit
rewards_margins	0.0428471	138	905	explicit
loss	0.974344	139	907	explicit
rewards_chosen	0.0824888	139	908	explicit
rewards_rejected	-0.0222759	139	909	explicit
rewards_accuracies	0.5625	139	910	explicit
rewards_margins	0.104765	139	911	explicit
loss	0.934074	140	913	explicit
rewards_chosen	0.0645451	140	914	explicit
rewards_rejected	-0.208322	140	915	explicit
rewards_accuracies	0.6875	140	916	explicit
rewards_margins	0.272867	140	917	explicit
loss	0.979175	141	919	explicit
rewards_chosen	-0.024251	141	920	explicit
rewards_rejected	-0.109159	141	921	explicit
rewards_accuracies	0.75	141	922	explicit
rewards_margins	0.0849079	141	923	explicit
loss	0.960242	142	925	explicit
rewards_chosen	0.0886326	142	926	explicit
rewards_rejected	-0.0739594	142	927	explicit
rewards_accuracies	0.5	142	928	explicit
rewards_margins	0.162592	142	929	explicit
loss	0.9365	143	931	explicit
rewards_chosen	0.0806057	143	932	explicit
rewards_rejected	-0.180141	143	933	explicit
rewards_accuracies	0.6875	143	934	explicit
rewards_margins	0.260746	143	935	explicit
loss	0.932613	144	937	explicit
rewards_chosen	0.104411	144	938	explicit
rewards_rejected	-0.178439	144	939	explicit
rewards_accuracies	0.6875	144	940	explicit
rewards_margins	0.28285	144	941	explicit
loss	0.982335	145	943	explicit
rewards_chosen	0.0934269	145	944	explicit
rewards_rejected	0.0195969	145	945	explicit
rewards_accuracies	0.3125	145	946	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.07383	145	947	explicit
loss	0.966703	146	949	explicit
rewards_chosen	0.0629463	146	950	explicit
rewards_rejected	-0.0739998	146	951	explicit
rewards_accuracies	0.5625	146	952	explicit
rewards_margins	0.136946	146	953	explicit
loss	0.925318	147	955	explicit
rewards_chosen	0.0721985	147	956	explicit
rewards_rejected	-0.234709	147	957	explicit
rewards_accuracies	0.6875	147	958	explicit
rewards_margins	0.306908	147	959	explicit
loss	0.966391	148	961	explicit
rewards_chosen	0.232932	148	962	explicit
rewards_rejected	0.0934998	148	963	explicit
rewards_accuracies	0.5625	148	964	explicit
rewards_margins	0.139433	148	965	explicit
loss	0.974617	149	967	explicit
rewards_chosen	-0.0340048	149	968	explicit
rewards_rejected	-0.13896	149	969	explicit
rewards_accuracies	0.5625	149	970	explicit
rewards_margins	0.104956	149	971	explicit
loss	0.958873	150	973	explicit
rewards_chosen	0.143062	150	974	explicit
rewards_rejected	-0.0236866	150	975	explicit
rewards_accuracies	0.6875	150	976	explicit
rewards_margins	0.166749	150	977	explicit
eval_rewards_chosen	0.0359891	150	979	explicit
eval_rewards_rejected	-0.146661	150	980	explicit
eval_rewards_accuracies	0.619444	150	981	explicit
eval_rewards_margins	0.18265	150	982	explicit
eval_loss	0.955486	150	984	explicit
eval_runtime	77.2216	150	985	explicit
eval_samples_per_second	4.662	150	986	explicit
eval_steps_per_second	0.583	150	987	explicit
eval_entropy	0.893641	150	988	explicit
eval_num_tokens	9.12418e+06	150	989	explicit
eval_logits_chosen	10.2916	150	990	explicit
eval_logits_rejected	10.3113	150	991	explicit
eval_mean_token_accuracy	0.850227	150	992	explicit
eval_rewards_chosen	0.0359891	150	993	explicit
eval_rewards_rejected	-0.146661	150	994	explicit
eval_rewards_accuracies	0.619444	150	995	explicit
eval_rewards_margins	0.18265	150	996	explicit
eval_logps_chosen	-1111.99	150	997	explicit
eval_logps_rejected	-1112.05	150	998	explicit
epoch	0.740741	150	999	explicit
phi	0.847792	150	1000	explicit
phi_bam	0.947662	150	1001	explicit
phi_bmq	0.788389	150	1002	explicit
phi_mku	0.874554	150	1003	explicit

Metric	Value	Step	Line	Source
phi_myk	0.858518	150	1004	explicit
phi_ses	0.673515	150	1005	explicit
phi_dtm	0.875799	150	1006	explicit
phi_snk	0.839542	150	1007	explicit
phi_boz	0.880553	150	1008	explicit
phi_ful	0.872129	150	1009	explicit
phi_spp	0.999123	150	1010	explicit
phi_taq	0.715931	150	1011	explicit
loss	0.900604	151	1012	explicit
rewards_chosen	0.108329	151	1013	explicit
rewards_rejected	-0.300475	151	1014	explicit
rewards_accuracies	0.875	151	1015	explicit
rewards_margins	0.408804	151	1016	explicit
loss	0.907681	152	1018	explicit
rewards_chosen	0.22069	152	1019	explicit
rewards_rejected	-0.16573	152	1020	explicit
rewards_accuracies	0.625	152	1021	explicit
rewards_margins	0.386421	152	1022	explicit
loss	0.95516	153	1024	explicit
rewards_chosen	-0.05896	153	1025	explicit
rewards_rejected	-0.243448	153	1026	explicit
rewards_accuracies	0.75	153	1027	explicit
rewards_margins	0.184488	153	1028	explicit
loss	0.906233	154	1030	explicit
rewards_chosen	0.146579	154	1031	explicit
rewards_rejected	-0.242405	154	1032	explicit
rewards_accuracies	0.625	154	1033	explicit
rewards_margins	0.388984	154	1034	explicit
loss	0.927244	155	1036	explicit
rewards_chosen	0.121647	155	1037	explicit
rewards_rejected	-0.175346	155	1038	explicit
rewards_accuracies	0.75	155	1039	explicit
rewards_margins	0.296992	155	1040	explicit
loss	0.960457	156	1042	explicit
rewards_chosen	0.0546158	156	1043	explicit
rewards_rejected	-0.108962	156	1044	explicit
rewards_accuracies	0.625	156	1045	explicit
rewards_margins	0.163578	156	1046	explicit
loss	0.95248	157	1048	explicit
rewards_chosen	0.253359	157	1049	explicit
rewards_rejected	0.0588409	157	1050	explicit
rewards_accuracies	0.75	157	1051	explicit
rewards_margins	0.194518	157	1052	explicit
loss	1.00054	158	1054	explicit
rewards_chosen	0.0472326	158	1055	explicit
rewards_rejected	0.0483138	158	1056	explicit
rewards_accuracies	0.3125	158	1057	explicit
rewards_margins	-0.00108118	158	1058	explicit
loss	0.925185	159	1060	explicit
rewards_chosen	0.197784	159	1061	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.113618	159	1062	explicit
rewards_accuracies	0.625	159	1063	explicit
rewards_margins	0.311402	159	1064	explicit
loss	0.941258	160	1066	explicit
rewards_chosen	0.252192	160	1067	explicit
rewards_rejected	0.00921937	160	1068	explicit
rewards_accuracies	0.6875	160	1069	explicit
rewards_margins	0.242972	160	1070	explicit
loss	0.949701	161	1072	explicit
rewards_chosen	0.256804	161	1073	explicit
rewards_rejected	0.0510328	161	1074	explicit
rewards_accuracies	0.625	161	1075	explicit
rewards_margins	0.205771	161	1076	explicit
loss	0.939565	162	1078	explicit
rewards_chosen	0.0697346	162	1079	explicit
rewards_rejected	-0.177244	162	1080	explicit
rewards_accuracies	0.625	162	1081	explicit
rewards_margins	0.246978	162	1082	explicit
loss	0.969771	163	1084	explicit
rewards_chosen	-0.0491882	163	1085	explicit
rewards_rejected	-0.172939	163	1086	explicit
rewards_accuracies	0.5625	163	1087	explicit
rewards_margins	0.123751	163	1088	explicit
loss	0.938819	164	1090	explicit
rewards_chosen	0.162262	164	1091	explicit
rewards_rejected	-0.0894424	164	1092	explicit
rewards_accuracies	0.5625	164	1093	explicit
rewards_margins	0.251705	164	1094	explicit
loss	0.98748	165	1096	explicit
rewards_chosen	0.10242	165	1097	explicit
rewards_rejected	0.0488724	165	1098	explicit
rewards_accuracies	0.625	165	1099	explicit
rewards_margins	0.0535477	165	1100	explicit
loss	0.94377	166	1102	explicit
rewards_chosen	0.162012	166	1103	explicit
rewards_rejected	-0.0673981	166	1104	explicit
rewards_accuracies	0.625	166	1105	explicit
rewards_margins	0.22941	166	1106	explicit
loss	0.943901	167	1108	explicit
rewards_chosen	0.0849934	167	1109	explicit
rewards_rejected	-0.146328	167	1110	explicit
rewards_accuracies	0.75	167	1111	explicit
rewards_margins	0.231321	167	1112	explicit
loss	0.972765	168	1114	explicit
rewards_chosen	0.130511	168	1115	explicit
rewards_rejected	0.0178581	168	1116	explicit
rewards_accuracies	0.4375	168	1117	explicit
rewards_margins	0.112653	168	1118	explicit
loss	0.961976	169	1120	explicit
rewards_chosen	-0.0609573	169	1121	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.220127	169	1122	explicit
rewards_accuracies	0.625	169	1123	explicit
rewards_margins	0.159169	169	1124	explicit
loss	0.931867	170	1126	explicit
rewards_chosen	0.165755	170	1127	explicit
rewards_rejected	-0.119616	170	1128	explicit
rewards_accuracies	0.625	170	1129	explicit
rewards_margins	0.285371	170	1130	explicit
loss	0.946966	171	1132	explicit
rewards_chosen	0.0863674	171	1133	explicit
rewards_rejected	-0.131672	171	1134	explicit
rewards_accuracies	0.625	171	1135	explicit
rewards_margins	0.218039	171	1136	explicit
loss	0.990036	172	1138	explicit
rewards_chosen	0.0372398	172	1139	explicit
rewards_rejected	-0.00227852	172	1140	explicit
rewards_accuracies	0.4375	172	1141	explicit
rewards_margins	0.0395184	172	1142	explicit
loss	0.94112	173	1144	explicit
rewards_chosen	0.122582	173	1145	explicit
rewards_rejected	-0.116703	173	1146	explicit
rewards_accuracies	0.8125	173	1147	explicit
rewards_margins	0.239285	173	1148	explicit
loss	0.941055	174	1150	explicit
rewards_chosen	0.0554729	174	1151	explicit
rewards_rejected	-0.191314	174	1152	explicit
rewards_accuracies	0.625	174	1153	explicit
rewards_margins	0.246787	174	1154	explicit
loss	0.961761	175	1156	explicit
rewards_chosen	0.0654358	175	1157	explicit
rewards_rejected	-0.0919697	175	1158	explicit
rewards_accuracies	0.625	175	1159	explicit
rewards_margins	0.157405	175	1160	explicit
loss	0.957447	176	1162	explicit
rewards_chosen	0.0167934	176	1163	explicit
rewards_rejected	-0.159979	176	1164	explicit
rewards_accuracies	0.5625	176	1165	explicit
rewards_margins	0.176772	176	1166	explicit
loss	0.917602	177	1168	explicit
rewards_chosen	0.0671581	177	1169	explicit
rewards_rejected	-0.28998	177	1170	explicit
rewards_accuracies	0.6875	177	1171	explicit
rewards_margins	0.357138	177	1172	explicit
loss	0.967728	178	1174	explicit
rewards_chosen	0.0596603	178	1175	explicit
rewards_rejected	-0.0728172	178	1176	explicit
rewards_accuracies	0.4375	178	1177	explicit
rewards_margins	0.132478	178	1178	explicit
loss	0.887555	179	1180	explicit
rewards_chosen	0.146545	179	1181	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.322552	179	1182	explicit
rewards_accuracies	0.875	179	1183	explicit
rewards_margins	0.469097	179	1184	explicit
loss	0.909081	180	1186	explicit
rewards_chosen	0.112595	180	1187	explicit
rewards_rejected	-0.264109	180	1188	explicit
rewards_accuracies	0.6875	180	1189	explicit
rewards_margins	0.376704	180	1190	explicit
loss	0.925053	181	1192	explicit
rewards_chosen	0.248834	181	1193	explicit
rewards_rejected	-0.068374	181	1194	explicit
rewards_accuracies	0.8125	181	1195	explicit
rewards_margins	0.317208	181	1196	explicit
loss	0.952353	182	1198	explicit
rewards_chosen	0.213041	182	1199	explicit
rewards_rejected	0.0182777	182	1200	explicit
rewards_accuracies	0.5625	182	1201	explicit
rewards_margins	0.194763	182	1202	explicit
loss	0.895801	183	1204	explicit
rewards_chosen	0.0203045	183	1205	explicit
rewards_rejected	-0.418752	183	1206	explicit
rewards_accuracies	0.75	183	1207	explicit
rewards_margins	0.439057	183	1208	explicit
loss	0.934727	184	1210	explicit
rewards_chosen	0.0019764	184	1211	explicit
rewards_rejected	-0.268652	184	1212	explicit
rewards_accuracies	0.8125	184	1213	explicit
rewards_margins	0.270628	184	1214	explicit
loss	0.943913	185	1216	explicit
rewards_chosen	0.0477753	185	1217	explicit
rewards_rejected	-0.182331	185	1218	explicit
rewards_accuracies	0.625	185	1219	explicit
rewards_margins	0.230107	185	1220	explicit
loss	0.879212	186	1222	explicit
rewards_chosen	0.160514	186	1223	explicit
rewards_rejected	-0.34407	186	1224	explicit
rewards_accuracies	0.875	186	1225	explicit
rewards_margins	0.504584	186	1226	explicit
loss	0.940834	187	1228	explicit
rewards_chosen	0.171703	187	1229	explicit
rewards_rejected	-0.0722198	187	1230	explicit
rewards_accuracies	0.75	187	1231	explicit
rewards_margins	0.243923	187	1232	explicit
loss	0.975078	188	1234	explicit
rewards_chosen	-0.0791164	188	1235	explicit
rewards_rejected	-0.18364	188	1236	explicit
rewards_accuracies	0.5	188	1237	explicit
rewards_margins	0.104524	188	1238	explicit
loss	0.903572	189	1240	explicit
rewards_chosen	0.137196	189	1241	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.266232	189	1242	explicit
rewards_accuracies	0.875	189	1243	explicit
rewards_margins	0.403428	189	1244	explicit
loss	0.944383	190	1246	explicit
rewards_chosen	0.11903	190	1247	explicit
rewards_rejected	-0.112641	190	1248	explicit
rewards_accuracies	0.8125	190	1249	explicit
rewards_margins	0.231671	190	1250	explicit
loss	0.939112	191	1252	explicit
rewards_chosen	-0.01077	191	1253	explicit
rewards_rejected	-0.261993	191	1254	explicit
rewards_accuracies	0.75	191	1255	explicit
rewards_margins	0.251223	191	1256	explicit
loss	0.984191	192	1258	explicit
rewards_chosen	-0.0790863	192	1259	explicit
rewards_rejected	-0.145096	192	1260	explicit
rewards_accuracies	0.625	192	1261	explicit
rewards_margins	0.0660099	192	1262	explicit
loss	0.8682	193	1264	explicit
rewards_chosen	0.246407	193	1265	explicit
rewards_rejected	-0.305497	193	1266	explicit
rewards_accuracies	0.875	193	1267	explicit
rewards_margins	0.551904	193	1268	explicit
loss	0.96413	194	1270	explicit
rewards_chosen	0.152963	194	1271	explicit
rewards_rejected	0.00514335	194	1272	explicit
rewards_accuracies	0.4375	194	1273	explicit
rewards_margins	0.14782	194	1274	explicit
loss	0.94506	195	1276	explicit
rewards_chosen	0.138061	195	1277	explicit
rewards_rejected	-0.0923714	195	1278	explicit
rewards_accuracies	0.5625	195	1279	explicit
rewards_margins	0.230432	195	1280	explicit
loss	0.93967	196	1282	explicit
rewards_chosen	0.160189	196	1283	explicit
rewards_rejected	-0.0945549	196	1284	explicit
rewards_accuracies	0.6875	196	1285	explicit
rewards_margins	0.254744	196	1286	explicit
loss	0.93933	197	1288	explicit
rewards_chosen	0.18557	197	1289	explicit
rewards_rejected	-0.0614934	197	1290	explicit
rewards_accuracies	0.75	197	1291	explicit
rewards_margins	0.247063	197	1292	explicit
loss	0.918304	198	1294	explicit
rewards_chosen	0.00539169	198	1295	explicit
rewards_rejected	-0.353874	198	1296	explicit
rewards_accuracies	0.6875	198	1297	explicit
rewards_margins	0.359266	198	1298	explicit
loss	0.944362	199	1300	explicit
rewards_chosen	0.269307	199	1301	explicit

Metric	Value	Step	Line	Source
rewards_rejected	0.0356545	199	1302	explicit
rewards_accuracies	0.625	199	1303	explicit
rewards_margins	0.233653	199	1304	explicit
loss	0.870843	200	1306	explicit
rewards_chosen	0.279268	200	1307	explicit
rewards_rejected	-0.26309	200	1308	explicit
rewards_accuracies	0.6875	200	1309	explicit
rewards_margins	0.542358	200	1310	explicit
eval_rewards_chosen	0.21471	200	1312	explicit
eval_rewards_rejected	-0.0493001	200	1313	explicit
eval_rewards_accuracies	0.630556	200	1314	explicit
eval_rewards_margins	0.26401	200	1315	explicit
eval_loss	0.936698	200	1317	explicit
eval_runtime	77.2064	200	1318	explicit
eval_samples_per_second	4.663	200	1319	explicit
eval_steps_per_second	0.583	200	1320	explicit
eval_entropy	0.888716	200	1321	explicit
eval_num_tokens	1.21923e+07	200	1322	explicit
eval_logits_chosen	10.2111	200	1323	explicit
eval_logits_rejected	10.2405	200	1324	explicit
eval_mean_token_accuracy	0.850604	200	1325	explicit
eval_rewards_chosen	0.21471	200	1326	explicit
eval_rewards_rejected	-0.0493001	200	1327	explicit
eval_rewards_accuracies	0.630556	200	1328	explicit
eval_rewards_margins	0.26401	200	1329	explicit
eval_logps_chosen	-1110.2	200	1330	explicit
eval_logps_rejected	-1111.08	200	1331	explicit
epoch	0.987654	200	1332	explicit
phi	0.847792	200	1333	explicit
phi_bam	0.947662	200	1334	explicit
phi_bmq	0.788389	200	1335	explicit
phi_mku	0.874554	200	1336	explicit
phi_myk	0.858518	200	1337	explicit
phi_ses	0.673515	200	1338	explicit
phi_dtm	0.875799	200	1339	explicit
phi_snk	0.839542	200	1340	explicit
phi_boz	0.880553	200	1341	explicit
phi_ful	0.872129	200	1342	explicit
phi_spp	0.999123	200	1343	explicit
phi_taq	0.715931	200	1344	explicit
loss	0.962713	201	1345	explicit
rewards_chosen	0.140826	201	1346	explicit
rewards_rejected	-0.0134741	201	1347	explicit
rewards_accuracies	0.5625	201	1348	explicit
rewards_margins	0.1543	201	1349	explicit
loss	0.912009	202	1351	explicit
rewards_chosen	0.332567	202	1352	explicit
rewards_rejected	-0.043264	202	1353	explicit
rewards_accuracies	0.875	202	1354	explicit
rewards_margins	0.375831	202	1355	explicit

Metric	Value	Step	Line	Source
loss	0.931014	203	1357	explicit
rewards_chosen	0.0892319	203	1358	explicit
rewards_rejected	-0.196209	203	1359	explicit
rewards_accuracies	0.75	203	1360	explicit
rewards_margins	0.285441	203	1361	explicit
loss	0.889384	204	1363	explicit
rewards_chosen	0.241437	204	1364	explicit
rewards_rejected	-0.21646	204	1365	explicit
rewards_accuracies	0.75	204	1366	explicit
rewards_margins	0.457897	204	1367	explicit
loss	0.922619	205	1369	explicit
rewards_chosen	0.0793633	205	1370	explicit
rewards_rejected	-0.242357	205	1371	explicit
rewards_accuracies	0.6875	205	1372	explicit
rewards_margins	0.32172	205	1373	explicit
loss	0.883027	206	1375	explicit
rewards_chosen	0.362519	206	1376	explicit
rewards_rejected	-0.120814	206	1377	explicit
rewards_accuracies	0.875	206	1378	explicit
rewards_margins	0.483333	206	1379	explicit
loss	0.962496	207	1381	explicit
rewards_chosen	0.0911758	207	1382	explicit
rewards_rejected	-0.0596535	207	1383	explicit
rewards_accuracies	0.625	207	1384	explicit
rewards_margins	0.150829	207	1385	explicit
loss	0.940614	208	1387	explicit
rewards_chosen	0.209809	208	1388	explicit
rewards_rejected	-0.0433052	208	1389	explicit
rewards_accuracies	0.6875	208	1390	explicit
rewards_margins	0.253115	208	1391	explicit
loss	0.962112	209	1393	explicit
rewards_chosen	0.133706	209	1394	explicit
rewards_rejected	-0.033263	209	1395	explicit
rewards_accuracies	0.5	209	1396	explicit
rewards_margins	0.166969	209	1397	explicit
loss	0.933137	210	1399	explicit
rewards_chosen	0.177279	210	1400	explicit
rewards_rejected	-0.107918	210	1401	explicit
rewards_accuracies	0.8125	210	1402	explicit
rewards_margins	0.285197	210	1403	explicit
loss	0.931638	211	1405	explicit
rewards_chosen	0.167224	211	1406	explicit
rewards_rejected	-0.114722	211	1407	explicit
rewards_accuracies	0.8125	211	1408	explicit
rewards_margins	0.281946	211	1409	explicit
loss	0.914362	212	1411	explicit
rewards_chosen	0.208566	212	1412	explicit
rewards_rejected	-0.143029	212	1413	explicit
rewards_accuracies	0.6875	212	1414	explicit
rewards_margins	0.351594	212	1415	explicit

Metric	Value	Step	Line	Source
loss	0.921845	213	1417	explicit
rewards_chosen	0.0269215	213	1418	explicit
rewards_rejected	-0.306101	213	1419	explicit
rewards_accuracies	0.8125	213	1420	explicit
rewards_margins	0.333023	213	1421	explicit
loss	0.868431	214	1423	explicit
rewards_chosen	0.312773	214	1424	explicit
rewards_rejected	-0.244367	214	1425	explicit
rewards_accuracies	0.8125	214	1426	explicit
rewards_margins	0.55714	214	1427	explicit
loss	0.927261	215	1429	explicit
rewards_chosen	0.325265	215	1430	explicit
rewards_rejected	0.0152195	215	1431	explicit
rewards_accuracies	0.5625	215	1432	explicit
rewards_margins	0.310046	215	1433	explicit
loss	0.90245	216	1435	explicit
rewards_chosen	0.34022	216	1436	explicit
rewards_rejected	-0.0705021	216	1437	explicit
rewards_accuracies	0.625	216	1438	explicit
rewards_margins	0.410722	216	1439	explicit
loss	0.919576	217	1441	explicit
rewards_chosen	0.29238	217	1442	explicit
rewards_rejected	-0.035284	217	1443	explicit
rewards_accuracies	0.875	217	1444	explicit
rewards_margins	0.327664	217	1445	explicit
loss	0.910155	218	1447	explicit
rewards_chosen	0.336204	218	1448	explicit
rewards_rejected	-0.0510224	218	1449	explicit
rewards_accuracies	0.5625	218	1450	explicit
rewards_margins	0.387226	218	1451	explicit
loss	0.908488	219	1453	explicit
rewards_chosen	0.359203	219	1454	explicit
rewards_rejected	-0.048719	219	1455	explicit
rewards_accuracies	0.5	219	1456	explicit
rewards_margins	0.407922	219	1457	explicit
loss	0.915776	220	1459	explicit
rewards_chosen	0.347471	220	1460	explicit
rewards_rejected	0.00233088	220	1461	explicit
rewards_accuracies	0.6875	220	1462	explicit
rewards_margins	0.345141	220	1463	explicit
loss	0.862569	221	1465	explicit
rewards_chosen	0.392094	221	1466	explicit
rewards_rejected	-0.192302	221	1467	explicit
rewards_accuracies	0.6875	221	1468	explicit
rewards_margins	0.584396	221	1469	explicit
loss	0.921852	222	1471	explicit
rewards_chosen	0.199861	222	1472	explicit
rewards_rejected	-0.12488	222	1473	explicit
rewards_accuracies	0.8125	222	1474	explicit
rewards_margins	0.324741	222	1475	explicit

Metric	Value	Step	Line	Source
loss	0.957759	223	1477	explicit
rewards_chosen	0.223979	223	1478	explicit
rewards_rejected	0.0436078	223	1479	explicit
rewards_accuracies	0.5625	223	1480	explicit
rewards_margins	0.180371	223	1481	explicit
loss	0.844011	224	1483	explicit
rewards_chosen	0.440392	224	1484	explicit
rewards_rejected	-0.226907	224	1485	explicit
rewards_accuracies	0.8125	224	1486	explicit
rewards_margins	0.667299	224	1487	explicit
loss	0.946204	225	1489	explicit
rewards_chosen	0.044754	225	1490	explicit
rewards_rejected	-0.182604	225	1491	explicit
rewards_accuracies	0.5	225	1492	explicit
rewards_margins	0.227358	225	1493	explicit
loss	0.907042	226	1495	explicit
rewards_chosen	0.335318	226	1496	explicit
rewards_rejected	-0.0473938	226	1497	explicit
rewards_accuracies	0.8125	226	1498	explicit
rewards_margins	0.382712	226	1499	explicit
loss	0.940186	227	1501	explicit
rewards_chosen	0.176321	227	1502	explicit
rewards_rejected	-0.0817509	227	1503	explicit
rewards_accuracies	0.5	227	1504	explicit
rewards_margins	0.258072	227	1505	explicit
loss	0.905193	228	1507	explicit
rewards_chosen	0.331749	228	1508	explicit
rewards_rejected	-0.0865158	228	1509	explicit
rewards_accuracies	0.5625	228	1510	explicit
rewards_margins	0.418265	228	1511	explicit
loss	0.964999	229	1513	explicit
rewards_chosen	0.0103146	229	1514	explicit
rewards_rejected	-0.138647	229	1515	explicit
rewards_accuracies	0.5625	229	1516	explicit
rewards_margins	0.148962	229	1517	explicit
loss	0.90523	230	1519	explicit
rewards_chosen	0.293573	230	1520	explicit
rewards_rejected	-0.148427	230	1521	explicit
rewards_accuracies	0.625	230	1522	explicit
rewards_margins	0.442001	230	1523	explicit
loss	0.953253	231	1525	explicit
rewards_chosen	0.116657	231	1526	explicit
rewards_rejected	-0.0922821	231	1527	explicit
rewards_accuracies	0.625	231	1528	explicit
rewards_margins	0.208939	231	1529	explicit
loss	0.928931	232	1531	explicit
rewards_chosen	0.121471	232	1532	explicit
rewards_rejected	-0.193365	232	1533	explicit
rewards_accuracies	0.75	232	1534	explicit
rewards_margins	0.314837	232	1535	explicit

Metric	Value	Step	Line	Source
loss	0.952967	233	1537	explicit
rewards_chosen	0.381271	233	1538	explicit
rewards_rejected	0.191503	233	1539	explicit
rewards_accuracies	0.5625	233	1540	explicit
rewards_margins	0.189768	233	1541	explicit
loss	0.934135	234	1543	explicit
rewards_chosen	0.106099	234	1544	explicit
rewards_rejected	-0.174501	234	1545	explicit
rewards_accuracies	0.625	234	1546	explicit
rewards_margins	0.2806	234	1547	explicit
loss	0.881718	235	1549	explicit
rewards_chosen	0.295264	235	1550	explicit
rewards_rejected	-0.204786	235	1551	explicit
rewards_accuracies	0.75	235	1552	explicit
rewards_margins	0.50005	235	1553	explicit
loss	0.928596	236	1555	explicit
rewards_chosen	0.160576	236	1556	explicit
rewards_rejected	-0.137373	236	1557	explicit
rewards_accuracies	0.75	236	1558	explicit
rewards_margins	0.297949	236	1559	explicit
loss	0.933981	237	1561	explicit
rewards_chosen	0.224352	237	1562	explicit
rewards_rejected	-0.0514044	237	1563	explicit
rewards_accuracies	0.75	237	1564	explicit
rewards_margins	0.275757	237	1565	explicit
loss	0.927821	238	1567	explicit
rewards_chosen	0.357777	238	1568	explicit
rewards_rejected	0.0564482	238	1569	explicit
rewards_accuracies	0.6875	238	1570	explicit
rewards_margins	0.301328	238	1571	explicit
loss	0.868547	239	1573	explicit
rewards_chosen	0.51903	239	1574	explicit
rewards_rejected	-0.0555267	239	1575	explicit
rewards_accuracies	0.75	239	1576	explicit
rewards_margins	0.574557	239	1577	explicit
loss	0.928479	240	1579	explicit
rewards_chosen	0.167424	240	1580	explicit
rewards_rejected	-0.139567	240	1581	explicit
rewards_accuracies	0.5625	240	1582	explicit
rewards_margins	0.30699	240	1583	explicit
loss	0.912332	241	1585	explicit
rewards_chosen	0.244066	241	1586	explicit
rewards_rejected	-0.124665	241	1587	explicit
rewards_accuracies	0.8125	241	1588	explicit
rewards_margins	0.368732	241	1589	explicit
loss	0.912173	242	1591	explicit
rewards_chosen	0.305503	242	1592	explicit
rewards_rejected	-0.0722401	242	1593	explicit
rewards_accuracies	0.6875	242	1594	explicit
rewards_margins	0.377743	242	1595	explicit

Metric	Value	Step	Line	Source
loss	0.910127	243	1597	explicit
rewards_chosen	0.314428	243	1598	explicit
rewards_rejected	-0.0743717	243	1599	explicit
rewards_accuracies	0.6875	243	1600	explicit
rewards_margins	0.3888	243	1601	explicit
loss	0.927207	244	1603	explicit
rewards_chosen	0.15168	244	1604	explicit
rewards_rejected	-0.162738	244	1605	explicit
rewards_accuracies	0.6875	244	1606	explicit
rewards_margins	0.314418	244	1607	explicit
loss	0.910616	245	1609	explicit
rewards_chosen	0.210691	245	1610	explicit
rewards_rejected	-0.165788	245	1611	explicit
rewards_accuracies	0.8125	245	1612	explicit
rewards_margins	0.37648	245	1613	explicit
loss	0.865798	246	1615	explicit
rewards_chosen	0.477171	246	1616	explicit
rewards_rejected	-0.0867861	246	1617	explicit
rewards_accuracies	0.75	246	1618	explicit
rewards_margins	0.563957	246	1619	explicit
loss	0.931443	247	1621	explicit
rewards_chosen	0.282407	247	1622	explicit
rewards_rejected	-0.00837073	247	1623	explicit
rewards_accuracies	0.75	247	1624	explicit
rewards_margins	0.290778	247	1625	explicit
loss	0.936488	248	1627	explicit
rewards_chosen	0.473677	248	1628	explicit
rewards_rejected	0.205566	248	1629	explicit
rewards_accuracies	0.4375	248	1630	explicit
rewards_margins	0.268111	248	1631	explicit
loss	0.852273	249	1633	explicit
rewards_chosen	0.410458	249	1634	explicit
rewards_rejected	-0.213524	249	1635	explicit
rewards_accuracies	0.9375	249	1636	explicit
rewards_margins	0.623982	249	1637	explicit
loss	0.886829	250	1639	explicit
rewards_chosen	0.394995	250	1640	explicit
rewards_rejected	-0.105366	250	1641	explicit
rewards_accuracies	0.5625	250	1642	explicit
rewards_margins	0.500361	250	1643	explicit
eval_rewards_chosen	0.31604	250	1645	explicit
eval_rewards_rejected	-0.0502559	250	1646	explicit
eval_rewards_accuracies	0.661111	250	1647	explicit
eval_rewards_margins	0.366296	250	1648	explicit
eval_loss	0.914309	250	1650	explicit
eval_runtime	77.2372	250	1651	explicit
eval_samples_per_second	4.661	250	1652	explicit
eval_steps_per_second	0.583	250	1653	explicit
eval_entropy	0.886195	250	1654	explicit
eval_num_tokens	1.52077e+07	250	1655	explicit

Metric	Value	Step	Line	Source
eval_logits_chosen	10.1435	250	1656	explicit
eval_logits_rejected	10.1696	250	1657	explicit
eval_mean_token_accuracy	0.850712	250	1658	explicit
eval_rewards_chosen	0.31604	250	1659	explicit
eval_rewards_rejected	-0.0502559	250	1660	explicit
eval_rewards_accuracies	0.661111	250	1661	explicit
eval_rewards_margins	0.366296	250	1662	explicit
eval_logps_chosen	-1109.19	250	1663	explicit
eval_logps_rejected	-1111.09	250	1664	explicit
epoch	1.2321	250	1665	explicit
phi	0.847792	250	1666	explicit
phi_bam	0.947662	250	1667	explicit
phi_bmq	0.788389	250	1668	explicit
phi_mku	0.874554	250	1669	explicit
phi_myk	0.858518	250	1670	explicit
phi_ses	0.673515	250	1671	explicit
phi_dtm	0.875799	250	1672	explicit
phi_snk	0.839542	250	1673	explicit
phi_boz	0.880553	250	1674	explicit
phi_ful	0.872129	250	1675	explicit
phi_spp	0.999123	250	1676	explicit
phi_taq	0.715931	250	1677	explicit
loss	0.859028	251	1678	explicit
rewards_chosen	0.482402	251	1679	explicit
rewards_rejected	-0.11969	251	1680	explicit
rewards_accuracies	0.6875	251	1681	explicit
rewards_margins	0.602092	251	1682	explicit
loss	0.898851	252	1684	explicit
rewards_chosen	0.187127	252	1685	explicit
rewards_rejected	-0.242192	252	1686	explicit
rewards_accuracies	0.6875	252	1687	explicit
rewards_margins	0.429319	252	1688	explicit
loss	0.955172	253	1690	explicit
rewards_chosen	0.129077	253	1691	explicit
rewards_rejected	-0.0596252	253	1692	explicit
rewards_accuracies	0.625	253	1693	explicit
rewards_margins	0.188702	253	1694	explicit
loss	0.917352	254	1696	explicit
rewards_chosen	0.349191	254	1697	explicit
rewards_rejected	-0.00926218	254	1698	explicit
rewards_accuracies	0.75	254	1699	explicit
rewards_margins	0.358453	254	1700	explicit
loss	0.904621	255	1702	explicit
rewards_chosen	0.152782	255	1703	explicit
rewards_rejected	-0.273989	255	1704	explicit
rewards_accuracies	0.6875	255	1705	explicit
rewards_margins	0.426771	255	1706	explicit
loss	0.932793	256	1708	explicit
rewards_chosen	0.0879808	256	1709	explicit
rewards_rejected	-0.194153	256	1710	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.5625	256	1711	explicit
rewards_margins	0.282134	256	1712	explicit
loss	0.930619	257	1714	explicit
rewards_chosen	0.494393	257	1715	explicit
rewards_rejected	0.177692	257	1716	explicit
rewards_accuracies	0.8125	257	1717	explicit
rewards_margins	0.316701	257	1718	explicit
loss	0.942967	258	1720	explicit
rewards_chosen	0.202705	258	1721	explicit
rewards_rejected	-0.0353634	258	1722	explicit
rewards_accuracies	0.5625	258	1723	explicit
rewards_margins	0.238068	258	1724	explicit
loss	0.914377	259	1726	explicit
rewards_chosen	0.319444	259	1727	explicit
rewards_rejected	-0.0424072	259	1728	explicit
rewards_accuracies	0.5625	259	1729	explicit
rewards_margins	0.361851	259	1730	explicit
loss	0.908766	260	1732	explicit
rewards_chosen	0.357326	260	1733	explicit
rewards_rejected	-0.0230463	260	1734	explicit
rewards_accuracies	0.8125	260	1735	explicit
rewards_margins	0.380372	260	1736	explicit
loss	0.905122	261	1738	explicit
rewards_chosen	0.275782	261	1739	explicit
rewards_rejected	-0.129389	261	1740	explicit
rewards_accuracies	0.8125	261	1741	explicit
rewards_margins	0.405171	261	1742	explicit
loss	0.931242	262	1744	explicit
rewards_chosen	0.29877	262	1745	explicit
rewards_rejected	-0.00910568	262	1746	explicit
rewards_accuracies	0.6875	262	1747	explicit
rewards_margins	0.307876	262	1748	explicit
loss	0.913258	263	1750	explicit
rewards_chosen	0.382576	263	1751	explicit
rewards_rejected	0.012751	263	1752	explicit
rewards_accuracies	0.5625	263	1753	explicit
rewards_margins	0.369825	263	1754	explicit
loss	0.936742	264	1756	explicit
rewards_chosen	0.147879	264	1757	explicit
rewards_rejected	-0.110392	264	1758	explicit
rewards_accuracies	0.875	264	1759	explicit
rewards_margins	0.258272	264	1760	explicit
loss	0.874869	265	1762	explicit
rewards_chosen	0.18356	265	1763	explicit
rewards_rejected	-0.348544	265	1764	explicit
rewards_accuracies	0.875	265	1765	explicit
rewards_margins	0.532104	265	1766	explicit
loss	0.89586	266	1768	explicit
rewards_chosen	0.234052	266	1769	explicit
rewards_rejected	-0.22667	266	1770	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.8125	266	1771	explicit
rewards_margins	0.460722	266	1772	explicit
loss	0.925657	267	1774	explicit
rewards_chosen	0.395474	267	1775	explicit
rewards_rejected	0.0655872	267	1776	explicit
rewards_accuracies	0.6875	267	1777	explicit
rewards_margins	0.329887	267	1778	explicit
loss	0.857134	268	1780	explicit
rewards_chosen	0.448612	268	1781	explicit
rewards_rejected	-0.162629	268	1782	explicit
rewards_accuracies	0.875	268	1783	explicit
rewards_margins	0.611241	268	1784	explicit
loss	0.934737	269	1786	explicit
rewards_chosen	0.258418	269	1787	explicit
rewards_rejected	-0.0198681	269	1788	explicit
rewards_accuracies	0.5625	269	1789	explicit
rewards_margins	0.278286	269	1790	explicit
loss	0.852222	270	1792	explicit
rewards_chosen	0.135725	270	1793	explicit
rewards_rejected	-0.509574	270	1794	explicit
rewards_accuracies	0.6875	270	1795	explicit
rewards_margins	0.645299	270	1796	explicit
loss	0.899758	271	1798	explicit
rewards_chosen	0.271471	271	1799	explicit
rewards_rejected	-0.151517	271	1800	explicit
rewards_accuracies	0.8125	271	1801	explicit
rewards_margins	0.422989	271	1802	explicit
loss	0.930311	272	1804	explicit
rewards_chosen	0.20389	272	1805	explicit
rewards_rejected	-0.0958855	272	1806	explicit
rewards_accuracies	0.5625	272	1807	explicit
rewards_margins	0.299775	272	1808	explicit
loss	0.891263	273	1810	explicit
rewards_chosen	0.421977	273	1811	explicit
rewards_rejected	-0.0406769	273	1812	explicit
rewards_accuracies	0.625	273	1813	explicit
rewards_margins	0.462654	273	1814	explicit
loss	0.882744	274	1816	explicit
rewards_chosen	0.104836	274	1817	explicit
rewards_rejected	-0.39868	274	1818	explicit
rewards_accuracies	0.875	274	1819	explicit
rewards_margins	0.503516	274	1820	explicit
loss	0.909658	275	1822	explicit
rewards_chosen	0.254369	275	1823	explicit
rewards_rejected	-0.12889	275	1824	explicit
rewards_accuracies	0.6875	275	1825	explicit
rewards_margins	0.383259	275	1826	explicit
loss	0.878559	276	1828	explicit
rewards_chosen	0.285587	276	1829	explicit
rewards_rejected	-0.239376	276	1830	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.875	276	1831	explicit
rewards_margins	0.524963	276	1832	explicit
loss	0.87899	277	1834	explicit
rewards_chosen	0.327596	277	1835	explicit
rewards_rejected	-0.182243	277	1836	explicit
rewards_accuracies	0.6875	277	1837	explicit
rewards_margins	0.50984	277	1838	explicit
loss	0.897032	278	1840	explicit
rewards_chosen	0.380618	278	1841	explicit
rewards_rejected	-0.0518871	278	1842	explicit
rewards_accuracies	0.6875	278	1843	explicit
rewards_margins	0.432505	278	1844	explicit
loss	0.963822	279	1846	explicit
rewards_chosen	0.169625	279	1847	explicit
rewards_rejected	0.0204294	279	1848	explicit
rewards_accuracies	0.625	279	1849	explicit
rewards_margins	0.149196	279	1850	explicit
loss	0.916447	280	1852	explicit
rewards_chosen	0.195565	280	1853	explicit
rewards_rejected	-0.162469	280	1854	explicit
rewards_accuracies	0.625	280	1855	explicit
rewards_margins	0.358035	280	1856	explicit
loss	0.885658	281	1858	explicit
rewards_chosen	0.401568	281	1859	explicit
rewards_rejected	-0.0992695	281	1860	explicit
rewards_accuracies	0.75	281	1861	explicit
rewards_margins	0.500838	281	1862	explicit
loss	0.916281	282	1864	explicit
rewards_chosen	0.269766	282	1865	explicit
rewards_rejected	-0.0888407	282	1866	explicit
rewards_accuracies	0.875	282	1867	explicit
rewards_margins	0.358607	282	1868	explicit
loss	0.897305	283	1870	explicit
rewards_chosen	0.40911	283	1871	explicit
rewards_rejected	-0.0366321	283	1872	explicit
rewards_accuracies	0.75	283	1873	explicit
rewards_margins	0.445742	283	1874	explicit
loss	0.892639	284	1876	explicit
rewards_chosen	0.207726	284	1877	explicit
rewards_rejected	-0.257753	284	1878	explicit
rewards_accuracies	0.6875	284	1879	explicit
rewards_margins	0.465479	284	1880	explicit
loss	0.884862	285	1882	explicit
rewards_chosen	0.227912	285	1883	explicit
rewards_rejected	-0.256532	285	1884	explicit
rewards_accuracies	0.8125	285	1885	explicit
rewards_margins	0.484444	285	1886	explicit
loss	0.905839	286	1888	explicit
rewards_chosen	0.329955	286	1889	explicit
rewards_rejected	-0.0615917	286	1890	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.5	286	1891	explicit
rewards_margins	0.391546	286	1892	explicit
loss	0.929622	287	1894	explicit
rewards_chosen	0.100825	287	1895	explicit
rewards_rejected	-0.188632	287	1896	explicit
rewards_accuracies	0.6875	287	1897	explicit
rewards_margins	0.289456	287	1898	explicit
loss	0.887535	288	1900	explicit
rewards_chosen	0.321547	288	1901	explicit
rewards_rejected	-0.162584	288	1902	explicit
rewards_accuracies	0.8125	288	1903	explicit
rewards_margins	0.484131	288	1904	explicit
loss	0.888627	289	1906	explicit
rewards_chosen	0.598755	289	1907	explicit
rewards_rejected	0.105772	289	1908	explicit
rewards_accuracies	0.5625	289	1909	explicit
rewards_margins	0.492983	289	1910	explicit
loss	0.876568	290	1912	explicit
rewards_chosen	0.281644	290	1913	explicit
rewards_rejected	-0.239216	290	1914	explicit
rewards_accuracies	0.75	290	1915	explicit
rewards_margins	0.52086	290	1916	explicit
loss	0.886491	291	1918	explicit
rewards_chosen	0.310063	291	1919	explicit
rewards_rejected	-0.180045	291	1920	explicit
rewards_accuracies	0.6875	291	1921	explicit
rewards_margins	0.490109	291	1922	explicit
loss	0.91021	292	1924	explicit
rewards_chosen	0.323909	292	1925	explicit
rewards_rejected	-0.0600212	292	1926	explicit
rewards_accuracies	0.625	292	1927	explicit
rewards_margins	0.38393	292	1928	explicit
loss	0.912315	293	1930	explicit
rewards_chosen	0.2027	293	1931	explicit
rewards_rejected	-0.178209	293	1932	explicit
rewards_accuracies	0.5625	293	1933	explicit
rewards_margins	0.380909	293	1934	explicit
loss	0.867911	294	1936	explicit
rewards_chosen	0.286758	294	1937	explicit
rewards_rejected	-0.278635	294	1938	explicit
rewards_accuracies	0.75	294	1939	explicit
rewards_margins	0.565393	294	1940	explicit
loss	0.91054	295	1942	explicit
rewards_chosen	0.317805	295	1943	explicit
rewards_rejected	-0.0653629	295	1944	explicit
rewards_accuracies	0.625	295	1945	explicit
rewards_margins	0.383168	295	1946	explicit
loss	0.882031	296	1948	explicit
rewards_chosen	0.3059	296	1949	explicit
rewards_rejected	-0.204085	296	1950	explicit

Metric	Value	Step	Line	Source
rewards_accuracies	0.75	296	1951	explicit
rewards_margins	0.509985	296	1952	explicit
loss	0.875471	297	1954	explicit
rewards_chosen	0.427798	297	1955	explicit
rewards_rejected	-0.11539	297	1956	explicit
rewards_accuracies	0.8125	297	1957	explicit
rewards_margins	0.543188	297	1958	explicit
loss	0.906962	298	1960	explicit
rewards_chosen	0.278613	298	1961	explicit
rewards_rejected	-0.112123	298	1962	explicit
rewards_accuracies	0.625	298	1963	explicit
rewards_margins	0.390736	298	1964	explicit
loss	0.883232	299	1966	explicit
rewards_chosen	0.473278	299	1967	explicit
rewards_rejected	-0.0228313	299	1968	explicit
rewards_accuracies	0.875	299	1969	explicit
rewards_margins	0.496109	299	1970	explicit
loss	0.932442	300	1972	explicit
rewards_chosen	0.431221	300	1973	explicit
rewards_rejected	0.148508	300	1974	explicit
rewards_accuracies	0.625	300	1975	explicit
rewards_margins	0.282713	300	1976	explicit
eval_rewards_chosen	0.335307	300	1978	explicit
eval_rewards_rejected	-0.0404922	300	1979	explicit
eval_rewards_accuracies	0.697222	300	1980	explicit
eval_rewards_margins	0.375799	300	1981	explicit
eval_loss	0.912257	300	1983	explicit
eval_runtime	77.2547	300	1984	explicit
eval_samples_per_second	4.66	300	1985	explicit
eval_steps_per_second	0.582	300	1986	explicit
eval_entropy	0.884795	300	1987	explicit
eval_num_tokens	1.82515e+07	300	1988	explicit
eval_logits_chosen	10.1254	300	1989	explicit
eval_logits_rejected	10.1603	300	1990	explicit
eval_mean_token_accuracy	0.850647	300	1991	explicit
eval_rewards_chosen	0.335307	300	1992	explicit
eval_rewards_rejected	-0.0404922	300	1993	explicit
eval_rewards_accuracies	0.697222	300	1994	explicit
eval_rewards_margins	0.375799	300	1995	explicit
eval_logps_chosen	-1109	300	1996	explicit
eval_logps_rejected	-1110.99	300	1997	explicit
epoch	1.47901	300	1998	explicit
phi	0.847792	300	1999	explicit
phi_bam	0.947662	300	2000	explicit
phi_bmq	0.788389	300	2001	explicit
phi_mku	0.874554	300	2002	explicit
phi_myk	0.858518	300	2003	explicit
phi_ses	0.673515	300	2004	explicit
phi_dtm	0.875799	300	2005	explicit
phi_snk	0.839542	300	2006	explicit

Metric	Value	Step	Line	Source
phi_boz	0.880553	300	2007	explicit
phi_ful	0.872129	300	2008	explicit
phi_spp	0.999123	300	2009	explicit
phi_taq	0.715931	300	2010	explicit
loss	0.917012	301	2011	explicit
rewards_chosen	0.178993	301	2012	explicit
rewards_rejected	-0.164072	301	2013	explicit
rewards_accuracies	0.6875	301	2014	explicit
rewards_margins	0.343066	301	2015	explicit
loss	0.894493	302	2017	explicit
rewards_chosen	0.430424	302	2018	explicit
rewards_rejected	-0.0208321	302	2019	explicit
rewards_accuracies	0.75	302	2020	explicit
rewards_margins	0.451256	302	2021	explicit
loss	0.903252	303	2023	explicit
rewards_chosen	0.176118	303	2024	explicit
rewards_rejected	-0.240972	303	2025	explicit
rewards_accuracies	0.625	303	2026	explicit
rewards_margins	0.417089	303	2027	explicit
loss	0.875357	304	2029	explicit
rewards_chosen	0.195515	304	2030	explicit
rewards_rejected	-0.335625	304	2031	explicit
rewards_accuracies	0.8125	304	2032	explicit
rewards_margins	0.53114	304	2033	explicit
loss	0.953965	305	2035	explicit
rewards_chosen	0.0629272	305	2036	explicit
rewards_rejected	-0.133554	305	2037	explicit
rewards_accuracies	0.75	305	2038	explicit
rewards_margins	0.196481	305	2039	explicit
loss	0.934227	306	2041	explicit
rewards_chosen	0.338765	306	2042	explicit
rewards_rejected	0.0566353	306	2043	explicit
rewards_accuracies	0.8125	306	2044	explicit
rewards_margins	0.282129	306	2045	explicit
loss	0.935039	307	2047	explicit
rewards_chosen	0.3223	307	2048	explicit
rewards_rejected	0.0515221	307	2049	explicit
rewards_accuracies	0.5625	307	2050	explicit
rewards_margins	0.270778	307	2051	explicit
loss	0.905362	308	2053	explicit
rewards_chosen	0.32752	308	2054	explicit
rewards_rejected	-0.0900948	308	2055	explicit
rewards_accuracies	0.75	308	2056	explicit
rewards_margins	0.417615	308	2057	explicit
loss	0.873325	309	2059	explicit
rewards_chosen	0.542467	309	2060	explicit
rewards_rejected	0.00271873	309	2061	explicit
rewards_accuracies	0.8125	309	2062	explicit
rewards_margins	0.539749	309	2063	explicit
loss	0.914153	310	2065	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.0657982	310	2066	explicit
rewards_rejected	-0.309797	310	2067	explicit
rewards_accuracies	0.5	310	2068	explicit
rewards_margins	0.375595	310	2069	explicit
loss	0.908974	311	2071	explicit
rewards_chosen	0.0467972	311	2072	explicit
rewards_rejected	-0.350856	311	2073	explicit
rewards_accuracies	0.8125	311	2074	explicit
rewards_margins	0.397653	311	2075	explicit
loss	0.940386	312	2077	explicit
rewards_chosen	0.383586	312	2078	explicit
rewards_rejected	0.132955	312	2079	explicit
rewards_accuracies	0.5	312	2080	explicit
rewards_margins	0.250631	312	2081	explicit
loss	0.88398	313	2083	explicit
rewards_chosen	0.287407	313	2084	explicit
rewards_rejected	-0.205478	313	2085	explicit
rewards_accuracies	0.875	313	2086	explicit
rewards_margins	0.492885	313	2087	explicit
loss	0.919543	314	2089	explicit
rewards_chosen	0.209737	314	2090	explicit
rewards_rejected	-0.130913	314	2091	explicit
rewards_accuracies	0.4375	314	2092	explicit
rewards_margins	0.34065	314	2093	explicit
loss	0.879941	315	2095	explicit
rewards_chosen	0.337837	315	2096	explicit
rewards_rejected	-0.197811	315	2097	explicit
rewards_accuracies	0.625	315	2098	explicit
rewards_margins	0.535648	315	2099	explicit
loss	0.884854	316	2101	explicit
rewards_chosen	0.226374	316	2102	explicit
rewards_rejected	-0.288351	316	2103	explicit
rewards_accuracies	0.6875	316	2104	explicit
rewards_margins	0.514725	316	2105	explicit
loss	0.915683	317	2107	explicit
rewards_chosen	0.224963	317	2108	explicit
rewards_rejected	-0.138775	317	2109	explicit
rewards_accuracies	0.6875	317	2110	explicit
rewards_margins	0.363739	317	2111	explicit
loss	0.946616	318	2113	explicit
rewards_chosen	0.206918	318	2114	explicit
rewards_rejected	-0.0147673	318	2115	explicit
rewards_accuracies	0.6875	318	2116	explicit
rewards_margins	0.221685	318	2117	explicit
loss	0.893564	319	2119	explicit
rewards_chosen	0.38697	319	2120	explicit
rewards_rejected	-0.07388	319	2121	explicit
rewards_accuracies	0.8125	319	2122	explicit
rewards_margins	0.46085	319	2123	explicit
loss	0.903831	320	2125	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.383218	320	2126	explicit
rewards_rejected	-0.0294041	320	2127	explicit
rewards_accuracies	0.875	320	2128	explicit
rewards_margins	0.412622	320	2129	explicit
loss	0.911693	321	2131	explicit
rewards_chosen	0.205674	321	2132	explicit
rewards_rejected	-0.192008	321	2133	explicit
rewards_accuracies	0.75	321	2134	explicit
rewards_margins	0.397682	321	2135	explicit
loss	0.841612	322	2137	explicit
rewards_chosen	0.437278	322	2138	explicit
rewards_rejected	-0.244092	322	2139	explicit
rewards_accuracies	0.8125	322	2140	explicit
rewards_margins	0.681369	322	2141	explicit
loss	0.900762	323	2143	explicit
rewards_chosen	0.452786	323	2144	explicit
rewards_rejected	0.0286785	323	2145	explicit
rewards_accuracies	0.4375	323	2146	explicit
rewards_margins	0.424108	323	2147	explicit
loss	0.933305	324	2149	explicit
rewards_chosen	0.217964	324	2150	explicit
rewards_rejected	-0.0668804	324	2151	explicit
rewards_accuracies	0.5	324	2152	explicit
rewards_margins	0.284844	324	2153	explicit
loss	0.83129	325	2155	explicit
rewards_chosen	0.439717	325	2156	explicit
rewards_rejected	-0.313792	325	2157	explicit
rewards_accuracies	0.75	325	2158	explicit
rewards_margins	0.753509	325	2159	explicit
loss	0.937536	326	2161	explicit
rewards_chosen	0.165666	326	2162	explicit
rewards_rejected	-0.0941728	326	2163	explicit
rewards_accuracies	0.625	326	2164	explicit
rewards_margins	0.259839	326	2165	explicit
loss	0.881126	327	2167	explicit
rewards_chosen	0.250171	327	2168	explicit
rewards_rejected	-0.296338	327	2169	explicit
rewards_accuracies	0.75	327	2170	explicit
rewards_margins	0.546508	327	2171	explicit
loss	0.917806	328	2173	explicit
rewards_chosen	0.169508	328	2174	explicit
rewards_rejected	-0.191483	328	2175	explicit
rewards_accuracies	0.75	328	2176	explicit
rewards_margins	0.360991	328	2177	explicit
loss	0.899358	329	2179	explicit
rewards_chosen	0.218524	329	2180	explicit
rewards_rejected	-0.207943	329	2181	explicit
rewards_accuracies	0.75	329	2182	explicit
rewards_margins	0.426467	329	2183	explicit
loss	0.932621	330	2185	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.240958	330	2186	explicit
rewards_rejected	-0.0442473	330	2187	explicit
rewards_accuracies	0.875	330	2188	explicit
rewards_margins	0.285205	330	2189	explicit
loss	0.906172	331	2191	explicit
rewards_chosen	0.497239	331	2192	explicit
rewards_rejected	0.0974022	331	2193	explicit
rewards_accuracies	0.6875	331	2194	explicit
rewards_margins	0.399837	331	2195	explicit
loss	0.916736	332	2197	explicit
rewards_chosen	0.376851	332	2198	explicit
rewards_rejected	0.0321056	332	2199	explicit
rewards_accuracies	0.625	332	2200	explicit
rewards_margins	0.344746	332	2201	explicit
loss	0.919242	333	2203	explicit
rewards_chosen	0.167017	333	2204	explicit
rewards_rejected	-0.167899	333	2205	explicit
rewards_accuracies	0.625	333	2206	explicit
rewards_margins	0.334916	333	2207	explicit
loss	0.852147	334	2209	explicit
rewards_chosen	0.29978	334	2210	explicit
rewards_rejected	-0.358857	334	2211	explicit
rewards_accuracies	0.75	334	2212	explicit
rewards_margins	0.658637	334	2213	explicit
loss	0.856123	335	2215	explicit
rewards_chosen	0.420077	335	2216	explicit
rewards_rejected	-0.201301	335	2217	explicit
rewards_accuracies	0.8125	335	2218	explicit
rewards_margins	0.621378	335	2219	explicit
loss	0.925086	336	2221	explicit
rewards_chosen	0.343708	336	2222	explicit
rewards_rejected	0.026112	336	2223	explicit
rewards_accuracies	0.8125	336	2224	explicit
rewards_margins	0.317596	336	2225	explicit
loss	0.871197	337	2227	explicit
rewards_chosen	0.466184	337	2228	explicit
rewards_rejected	-0.0929847	337	2229	explicit
rewards_accuracies	0.625	337	2230	explicit
rewards_margins	0.559168	337	2231	explicit
loss	0.962928	338	2233	explicit
rewards_chosen	0.202607	338	2234	explicit
rewards_rejected	0.0422997	338	2235	explicit
rewards_accuracies	0.625	338	2236	explicit
rewards_margins	0.160308	338	2237	explicit
loss	0.876972	339	2239	explicit
rewards_chosen	0.244285	339	2240	explicit
rewards_rejected	-0.267907	339	2241	explicit
rewards_accuracies	0.75	339	2242	explicit
rewards_margins	0.512192	339	2243	explicit
loss	0.912543	340	2245	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.345642	340	2246	explicit
rewards_rejected	-0.0219631	340	2247	explicit
rewards_accuracies	0.8125	340	2248	explicit
rewards_margins	0.367605	340	2249	explicit
loss	0.93263	341	2251	explicit
rewards_chosen	0.374462	341	2252	explicit
rewards_rejected	0.0922524	341	2253	explicit
rewards_accuracies	0.9375	341	2254	explicit
rewards_margins	0.28221	341	2255	explicit
loss	0.919669	342	2257	explicit
rewards_chosen	0.274607	342	2258	explicit
rewards_rejected	-0.0749059	342	2259	explicit
rewards_accuracies	0.75	342	2260	explicit
rewards_margins	0.349513	342	2261	explicit
loss	0.91859	343	2263	explicit
rewards_chosen	0.182969	343	2264	explicit
rewards_rejected	-0.160569	343	2265	explicit
rewards_accuracies	0.8125	343	2266	explicit
rewards_margins	0.343537	343	2267	explicit
loss	0.934451	344	2269	explicit
rewards_chosen	0.412919	344	2270	explicit
rewards_rejected	0.141138	344	2271	explicit
rewards_accuracies	0.8125	344	2272	explicit
rewards_margins	0.27178	344	2273	explicit
loss	0.882176	345	2275	explicit
rewards_chosen	0.280879	345	2276	explicit
rewards_rejected	-0.225814	345	2277	explicit
rewards_accuracies	0.875	345	2278	explicit
rewards_margins	0.506693	345	2279	explicit
loss	0.91241	346	2281	explicit
rewards_chosen	0.43521	346	2282	explicit
rewards_rejected	0.0482494	346	2283	explicit
rewards_accuracies	0.6875	346	2284	explicit
rewards_margins	0.386961	346	2285	explicit
loss	0.868871	347	2287	explicit
rewards_chosen	0.213107	347	2288	explicit
rewards_rejected	-0.354626	347	2289	explicit
rewards_accuracies	0.875	347	2290	explicit
rewards_margins	0.567733	347	2291	explicit
loss	0.895873	348	2293	explicit
rewards_chosen	0.435487	348	2294	explicit
rewards_rejected	-0.0109097	348	2295	explicit
rewards_accuracies	0.6875	348	2296	explicit
rewards_margins	0.446396	348	2297	explicit
loss	0.956193	349	2299	explicit
rewards_chosen	0.0539661	349	2300	explicit
rewards_rejected	-0.137552	349	2301	explicit
rewards_accuracies	0.6875	349	2302	explicit
rewards_margins	0.191518	349	2303	explicit
loss	0.923672	350	2305	explicit

Metric	Value	Step	Line	Source
rewards_chosen	0.194226	350	2306	explicit
rewards_rejected	-0.127515	350	2307	explicit
rewards_accuracies	0.625	350	2308	explicit
rewards_margins	0.321741	350	2309	explicit
eval_rewards_chosen	0.261481	350	2311	explicit
eval_rewards_rejected	-0.0927108	350	2312	explicit
eval_rewards_accuracies	0.672222	350	2313	explicit
eval_rewards_margins	0.354192	350	2314	explicit
eval_loss	0.91685	350	2316	explicit
eval_runtime	77.2765	350	2317	explicit
eval_samples_per_second	4.659	350	2318	explicit
eval_steps_per_second	0.582	350	2319	explicit
eval_entropy	0.88801	350	2320	explicit
eval_num_tokens	2.12943e+07	350	2321	explicit
eval_logits_chosen	10.1541	350	2322	explicit
eval_logits_rejected	10.1804	350	2323	explicit
eval_mean_token_accuracy	0.850561	350	2324	explicit
eval_rewards_chosen	0.261481	350	2325	explicit
eval_rewards_rejected	-0.0927108	350	2326	explicit
eval_rewards_accuracies	0.672222	350	2327	explicit
eval_rewards_margins	0.354192	350	2328	explicit
eval_logps_chosen	-1109.73	350	2329	explicit
eval_logps_rejected	-1111.51	350	2330	explicit
epoch	1.72593	350	2331	explicit
phi	0.847792	350	2332	explicit
phi_bam	0.947662	350	2333	explicit
phi_bmq	0.788389	350	2334	explicit
phi_mku	0.874554	350	2335	explicit
phi_myk	0.858518	350	2336	explicit
phi_ses	0.673515	350	2337	explicit
phi_dtm	0.875799	350	2338	explicit
phi_snk	0.839542	350	2339	explicit
phi_boz	0.880553	350	2340	explicit
phi_ful	0.872129	350	2341	explicit
phi_spp	0.999123	350	2342	explicit
phi_taq	0.715931	350	2343	explicit
loss	0.953708	351	2344	explicit
rewards_chosen	0.25074	351	2345	explicit
rewards_rejected	0.0574497	351	2346	explicit
rewards_accuracies	0.5	351	2347	explicit
rewards_margins	0.19329	351	2348	explicit
loss	0.903363	352	2350	explicit
rewards_chosen	0.299097	352	2351	explicit
rewards_rejected	-0.112216	352	2352	explicit
rewards_accuracies	0.6875	352	2353	explicit
rewards_margins	0.411313	352	2354	explicit
loss	0.931107	353	2356	explicit
rewards_chosen	0.152719	353	2357	explicit
rewards_rejected	-0.136385	353	2358	explicit
rewards_accuracies	0.5625	353	2359	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.289104	353	2360	explicit
loss	0.899582	354	2362	explicit
rewards_chosen	0.240654	354	2363	explicit
rewards_rejected	-0.182397	354	2364	explicit
rewards_accuracies	0.75	354	2365	explicit
rewards_margins	0.423051	354	2366	explicit
loss	0.90285	355	2368	explicit
rewards_chosen	0.143204	355	2369	explicit
rewards_rejected	-0.25954	355	2370	explicit
rewards_accuracies	0.75	355	2371	explicit
rewards_margins	0.402744	355	2372	explicit
loss	0.895678	356	2374	explicit
rewards_chosen	0.378766	356	2375	explicit
rewards_rejected	-0.0953003	356	2376	explicit
rewards_accuracies	0.75	356	2377	explicit
rewards_margins	0.474066	356	2378	explicit
loss	0.92431	357	2380	explicit
rewards_chosen	0.33457	357	2381	explicit
rewards_rejected	0.018367	357	2382	explicit
rewards_accuracies	0.6875	357	2383	explicit
rewards_margins	0.316203	357	2384	explicit
loss	0.942864	358	2386	explicit
rewards_chosen	0.263506	358	2387	explicit
rewards_rejected	0.0216862	358	2388	explicit
rewards_accuracies	0.75	358	2389	explicit
rewards_margins	0.24182	358	2390	explicit
loss	0.883512	359	2392	explicit
rewards_chosen	0.201366	359	2393	explicit
rewards_rejected	-0.288681	359	2394	explicit
rewards_accuracies	0.75	359	2395	explicit
rewards_margins	0.490047	359	2396	explicit
loss	0.955686	360	2398	explicit
rewards_chosen	0.382214	360	2399	explicit
rewards_rejected	0.190827	360	2400	explicit
rewards_accuracies	0.625	360	2401	explicit
rewards_margins	0.191387	360	2402	explicit
loss	0.894511	361	2404	explicit
rewards_chosen	0.211437	361	2405	explicit
rewards_rejected	-0.237292	361	2406	explicit
rewards_accuracies	0.625	361	2407	explicit
rewards_margins	0.448729	361	2408	explicit
loss	0.914967	362	2410	explicit
rewards_chosen	0.225509	362	2411	explicit
rewards_rejected	-0.130074	362	2412	explicit
rewards_accuracies	0.625	362	2413	explicit
rewards_margins	0.355583	362	2414	explicit
loss	0.929418	363	2416	explicit
rewards_chosen	0.35101	363	2417	explicit
rewards_rejected	0.0483667	363	2418	explicit
rewards_accuracies	0.5625	363	2419	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.302643	363	2420	explicit
loss	0.915923	364	2422	explicit
rewards_chosen	0.363671	364	2423	explicit
rewards_rejected	0.00673924	364	2424	explicit
rewards_accuracies	0.75	364	2425	explicit
rewards_margins	0.356932	364	2426	explicit
loss	0.906384	365	2428	explicit
rewards_chosen	0.477003	365	2429	explicit
rewards_rejected	0.0778388	365	2430	explicit
rewards_accuracies	0.6875	365	2431	explicit
rewards_margins	0.399165	365	2432	explicit
loss	0.805251	366	2434	explicit
rewards_chosen	0.505897	366	2435	explicit
rewards_rejected	-0.340532	366	2436	explicit
rewards_accuracies	0.875	366	2437	explicit
rewards_margins	0.846428	366	2438	explicit
loss	0.895704	367	2440	explicit
rewards_chosen	0.24	367	2441	explicit
rewards_rejected	-0.223928	367	2442	explicit
rewards_accuracies	0.8125	367	2443	explicit
rewards_margins	0.463928	367	2444	explicit
loss	0.916812	368	2446	explicit
rewards_chosen	0.109045	368	2447	explicit
rewards_rejected	-0.250603	368	2448	explicit
rewards_accuracies	0.8125	368	2449	explicit
rewards_margins	0.359649	368	2450	explicit
loss	0.893664	369	2452	explicit
rewards_chosen	0.272706	369	2453	explicit
rewards_rejected	-0.191049	369	2454	explicit
rewards_accuracies	0.9375	369	2455	explicit
rewards_margins	0.463754	369	2456	explicit
loss	0.884606	370	2458	explicit
rewards_chosen	0.277113	370	2459	explicit
rewards_rejected	-0.21061	370	2460	explicit
rewards_accuracies	0.625	370	2461	explicit
rewards_margins	0.487723	370	2462	explicit
loss	0.868001	371	2464	explicit
rewards_chosen	0.621158	371	2465	explicit
rewards_rejected	0.0530462	371	2466	explicit
rewards_accuracies	0.625	371	2467	explicit
rewards_margins	0.568112	371	2468	explicit
loss	0.890066	372	2470	explicit
rewards_chosen	-0.0044043	372	2471	explicit
rewards_rejected	-0.464983	372	2472	explicit
rewards_accuracies	0.75	372	2473	explicit
rewards_margins	0.460578	372	2474	explicit
loss	0.872003	373	2476	explicit
rewards_chosen	0.2595	373	2477	explicit
rewards_rejected	-0.281126	373	2478	explicit
rewards_accuracies	0.8125	373	2479	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.540627	373	2480	explicit
loss	0.862254	374	2482	explicit
rewards_chosen	0.170769	374	2483	explicit
rewards_rejected	-0.430276	374	2484	explicit
rewards_accuracies	0.75	374	2485	explicit
rewards_margins	0.601045	374	2486	explicit
loss	0.912926	375	2488	explicit
rewards_chosen	0.420052	375	2489	explicit
rewards_rejected	0.0431251	375	2490	explicit
rewards_accuracies	0.8125	375	2491	explicit
rewards_margins	0.376926	375	2492	explicit
loss	0.87187	376	2494	explicit
rewards_chosen	0.417075	376	2495	explicit
rewards_rejected	-0.153865	376	2496	explicit
rewards_accuracies	0.625	376	2497	explicit
rewards_margins	0.57094	376	2498	explicit
loss	0.822038	377	2500	explicit
rewards_chosen	0.612412	377	2501	explicit
rewards_rejected	-0.162005	377	2502	explicit
rewards_accuracies	1	377	2503	explicit
rewards_margins	0.774417	377	2504	explicit
loss	0.881464	378	2506	explicit
rewards_chosen	0.180884	378	2507	explicit
rewards_rejected	-0.354926	378	2508	explicit
rewards_accuracies	0.8125	378	2509	explicit
rewards_margins	0.53581	378	2510	explicit
loss	0.921777	379	2512	explicit
rewards_chosen	0.264651	379	2513	explicit
rewards_rejected	-0.0753734	379	2514	explicit
rewards_accuracies	0.6875	379	2515	explicit
rewards_margins	0.340024	379	2516	explicit
loss	0.900342	380	2518	explicit
rewards_chosen	0.24726	380	2519	explicit
rewards_rejected	-0.17714	380	2520	explicit
rewards_accuracies	0.625	380	2521	explicit
rewards_margins	0.4244	380	2522	explicit
loss	0.876359	381	2524	explicit
rewards_chosen	0.311704	381	2525	explicit
rewards_rejected	-0.22747	381	2526	explicit
rewards_accuracies	0.75	381	2527	explicit
rewards_margins	0.539174	381	2528	explicit
loss	0.892132	382	2530	explicit
rewards_chosen	0.419117	382	2531	explicit
rewards_rejected	-0.0588303	382	2532	explicit
rewards_accuracies	0.6875	382	2533	explicit
rewards_margins	0.477947	382	2534	explicit
loss	0.897756	383	2536	explicit
rewards_chosen	0.366088	383	2537	explicit
rewards_rejected	-0.0650707	383	2538	explicit
rewards_accuracies	0.875	383	2539	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.431158	383	2540	explicit
loss	0.889837	384	2542	explicit
rewards_chosen	0.405371	384	2543	explicit
rewards_rejected	-0.0787949	384	2544	explicit
rewards_accuracies	0.75	384	2545	explicit
rewards_margins	0.484166	384	2546	explicit
loss	0.837065	385	2548	explicit
rewards_chosen	0.487159	385	2549	explicit
rewards_rejected	-0.217373	385	2550	explicit
rewards_accuracies	0.9375	385	2551	explicit
rewards_margins	0.704532	385	2552	explicit
loss	0.869743	386	2554	explicit
rewards_chosen	0.377114	386	2555	explicit
rewards_rejected	-0.193553	386	2556	explicit
rewards_accuracies	0.5625	386	2557	explicit
rewards_margins	0.570667	386	2558	explicit
loss	0.956241	387	2560	explicit
rewards_chosen	0.137417	387	2561	explicit
rewards_rejected	-0.0456734	387	2562	explicit
rewards_accuracies	0.6875	387	2563	explicit
rewards_margins	0.18309	387	2564	explicit
loss	0.930545	388	2566	explicit
rewards_chosen	0.329635	388	2567	explicit
rewards_rejected	0.0395966	388	2568	explicit
rewards_accuracies	0.75	388	2569	explicit
rewards_margins	0.290038	388	2570	explicit
loss	0.896631	389	2572	explicit
rewards_chosen	0.344017	389	2573	explicit
rewards_rejected	-0.0954823	389	2574	explicit
rewards_accuracies	0.9375	389	2575	explicit
rewards_margins	0.439499	389	2576	explicit
loss	0.871308	390	2578	explicit
rewards_chosen	0.284957	390	2579	explicit
rewards_rejected	-0.273251	390	2580	explicit
rewards_accuracies	0.9375	390	2581	explicit
rewards_margins	0.558207	390	2582	explicit
loss	0.85211	391	2584	explicit
rewards_chosen	0.413349	391	2585	explicit
rewards_rejected	-0.224314	391	2586	explicit
rewards_accuracies	0.8125	391	2587	explicit
rewards_margins	0.637663	391	2588	explicit
loss	0.889703	392	2590	explicit
rewards_chosen	0.196553	392	2591	explicit
rewards_rejected	-0.271159	392	2592	explicit
rewards_accuracies	0.875	392	2593	explicit
rewards_margins	0.467712	392	2594	explicit
loss	0.865495	393	2596	explicit
rewards_chosen	0.332338	393	2597	explicit
rewards_rejected	-0.287342	393	2598	explicit
rewards_accuracies	0.625	393	2599	explicit

Metric	Value	Step	Line	Source
rewards_margins	0.61968	393	2600	explicit
loss	0.902167	394	2602	explicit
rewards_chosen	0.309855	394	2603	explicit
rewards_rejected	-0.108929	394	2604	explicit
rewards_accuracies	0.5625	394	2605	explicit
rewards_margins	0.418784	394	2606	explicit
loss	0.866733	395	2608	explicit
rewards_chosen	0.452256	395	2609	explicit
rewards_rejected	-0.114583	395	2610	explicit
rewards_accuracies	0.75	395	2611	explicit
rewards_margins	0.56684	395	2612	explicit
loss	0.909522	396	2614	explicit
rewards_chosen	0.368128	396	2615	explicit
rewards_rejected	-0.0186199	396	2616	explicit
rewards_accuracies	0.75	396	2617	explicit
rewards_margins	0.386748	396	2618	explicit
loss	0.962887	397	2620	explicit
rewards_chosen	0.331216	397	2621	explicit
rewards_rejected	0.178238	397	2622	explicit
rewards_accuracies	0.6875	397	2623	explicit
rewards_margins	0.152978	397	2624	explicit
loss	0.880164	398	2626	explicit
rewards_chosen	0.346975	398	2627	explicit
rewards_rejected	-0.166673	398	2628	explicit
rewards_accuracies	0.625	398	2629	explicit
rewards_margins	0.513647	398	2630	explicit
loss	0.923578	399	2632	explicit
rewards_chosen	0.333126	399	2633	explicit
rewards_rejected	0.00460206	399	2634	explicit
rewards_accuracies	0.5625	399	2635	explicit
rewards_margins	0.328524	399	2636	explicit
loss	0.879937	400	2638	explicit
rewards_chosen	0.306873	400	2639	explicit
rewards_rejected	-0.197817	400	2640	explicit
rewards_accuracies	0.75	400	2641	explicit
rewards_margins	0.50469	400	2642	explicit
eval_rewards_chosen	0.352666	400	2644	explicit
eval_rewards_rejected	-0.0205382	400	2645	explicit
eval_rewards_accuracies	0.702778	400	2646	explicit
eval_rewards_margins	0.373204	400	2647	explicit
eval_loss	0.912628	400	2649	explicit
eval_runtime	77.2802	400	2650	explicit
eval_samples_per_second	4.658	400	2651	explicit
eval_steps_per_second	0.582	400	2652	explicit
eval_entropy	0.884196	400	2653	explicit
eval_num_tokens	2.43551e+07	400	2654	explicit
eval_logits_chosen	10.1173	400	2655	explicit
eval_logits_rejected	10.1433	400	2656	explicit
eval_mean_token_accuracy	0.85072	400	2657	explicit
eval_rewards_chosen	0.352666	400	2658	explicit

Metric	Value	Step	Line	Source
eval_rewards_rejected	-0.0205382	400	2659	explicit
eval_rewards_accuracies	0.702778	400	2660	explicit
eval_rewards_margins	0.373204	400	2661	explicit
eval_logps_chosen	-1108.82	400	2662	explicit
eval_logps_rejected	-1110.79	400	2663	explicit
epoch	1.97284	400	2664	explicit
phi	0.847792	400	2665	explicit
phi_bam	0.947662	400	2666	explicit
phi_bmq	0.788389	400	2667	explicit
phi_mku	0.874554	400	2668	explicit
phi_myk	0.858518	400	2669	explicit
phi_ses	0.673515	400	2670	explicit
phi_dtm	0.875799	400	2671	explicit
phi_snk	0.839542	400	2672	explicit
phi_boz	0.880553	400	2673	explicit
phi_ful	0.872129	400	2674	explicit
phi_spp	0.999123	400	2675	explicit
phi_taq	0.715931	400	2676	explicit
loss	0.86823	401	2677	explicit
rewards_chosen	0.41998	401	2678	explicit
rewards_rejected	-0.14035	401	2679	explicit
rewards_accuracies	0.875	401	2680	explicit
rewards_margins	0.56033	401	2681	explicit
loss	0.950009	402	2683	explicit
rewards_chosen	0.0886974	402	2684	explicit
rewards_rejected	-0.118414	402	2685	explicit
rewards_accuracies	0.625	402	2686	explicit
rewards_margins	0.207111	402	2687	explicit
loss	0.923808	403	2689	explicit
rewards_chosen	0.133261	403	2690	explicit
rewards_rejected	-0.18804	403	2691	explicit
rewards_accuracies	0.6875	403	2692	explicit
rewards_margins	0.321301	403	2693	explicit
loss	0.899213	404	2695	explicit
rewards_chosen	0.353944	404	2696	explicit
rewards_rejected	-0.103386	404	2697	explicit
rewards_accuracies	0.6875	404	2698	explicit
rewards_margins	0.45733	404	2699	explicit
loss	0.919245	405	2701	explicit
rewards_chosen	0.284501	405	2702	explicit
rewards_rejected	-0.0636467	405	2703	explicit
rewards_accuracies	0.625	405	2704	explicit
rewards_margins	0.348148	405	2705	explicit
loss	0.944422	406	2707	explicit
rewards_chosen	0.460965	406	2708	explicit
rewards_rejected	0.222298	406	2709	explicit
rewards_accuracies	0.375	406	2710	explicit
rewards_margins	0.238667	406	2711	explicit
loss	0.843614	407	2713	explicit
rewards_chosen	0.375247	407	2714	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.284158	407	2715	explicit
rewards_accuracies	0.875	407	2716	explicit
rewards_margins	0.659405	407	2717	explicit
loss	0.907833	408	2719	explicit
rewards_chosen	0.326318	408	2720	explicit
rewards_rejected	-0.0578198	408	2721	explicit
rewards_accuracies	0.6875	408	2722	explicit
rewards_margins	0.384138	408	2723	explicit
loss	0.924674	409	2725	explicit
rewards_chosen	0.185553	409	2726	explicit
rewards_rejected	-0.161977	409	2727	explicit
rewards_accuracies	0.625	409	2728	explicit
rewards_margins	0.347531	409	2729	explicit
loss	0.91637	410	2731	explicit
rewards_chosen	0.564727	410	2732	explicit
rewards_rejected	0.206304	410	2733	explicit
rewards_accuracies	0.625	410	2734	explicit
rewards_margins	0.358422	410	2735	explicit
loss	0.860943	411	2737	explicit
rewards_chosen	0.346844	411	2738	explicit
rewards_rejected	-0.254505	411	2739	explicit
rewards_accuracies	0.75	411	2740	explicit
rewards_margins	0.601349	411	2741	explicit
loss	0.890549	412	2743	explicit
rewards_chosen	0.236769	412	2744	explicit
rewards_rejected	-0.239834	412	2745	explicit
rewards_accuracies	0.75	412	2746	explicit
rewards_margins	0.476602	412	2747	explicit
loss	0.868848	413	2749	explicit
rewards_chosen	0.146503	413	2750	explicit
rewards_rejected	-0.418321	413	2751	explicit
rewards_accuracies	0.75	413	2752	explicit
rewards_margins	0.564824	413	2753	explicit
loss	0.88476	414	2755	explicit
rewards_chosen	0.21805	414	2756	explicit
rewards_rejected	-0.268075	414	2757	explicit
rewards_accuracies	0.625	414	2758	explicit
rewards_margins	0.486125	414	2759	explicit
loss	0.939071	415	2761	explicit
rewards_chosen	0.310329	415	2762	explicit
rewards_rejected	0.0527805	415	2763	explicit
rewards_accuracies	0.5	415	2764	explicit
rewards_margins	0.257548	415	2765	explicit
loss	0.901963	416	2767	explicit
rewards_chosen	0.0968002	416	2768	explicit
rewards_rejected	-0.316595	416	2769	explicit
rewards_accuracies	0.875	416	2770	explicit
rewards_margins	0.413395	416	2771	explicit
loss	0.934938	417	2773	explicit
rewards_chosen	0.347257	417	2774	explicit

Metric	Value	Step	Line	Source
rewards_rejected	0.0688305	417	2775	explicit
rewards_accuracies	0.4375	417	2776	explicit
rewards_margins	0.278426	417	2777	explicit
loss	0.949511	418	2779	explicit
rewards_chosen	0.233146	418	2780	explicit
rewards_rejected	0.0233593	418	2781	explicit
rewards_accuracies	0.625	418	2782	explicit
rewards_margins	0.209786	418	2783	explicit
loss	0.889673	419	2785	explicit
rewards_chosen	0.37271	419	2786	explicit
rewards_rejected	-0.0956749	419	2787	explicit
rewards_accuracies	0.75	419	2788	explicit
rewards_margins	0.468385	419	2789	explicit
loss	0.934606	420	2791	explicit
rewards_chosen	0.305619	420	2792	explicit
rewards_rejected	0.0203903	420	2793	explicit
rewards_accuracies	0.5625	420	2794	explicit
rewards_margins	0.285228	420	2795	explicit
loss	0.962464	421	2797	explicit
rewards_chosen	0.274429	421	2798	explicit
rewards_rejected	0.117919	421	2799	explicit
rewards_accuracies	0.6875	421	2800	explicit
rewards_margins	0.15651	421	2801	explicit
loss	0.910767	422	2803	explicit
rewards_chosen	0.344636	422	2804	explicit
rewards_rejected	-0.0380878	422	2805	explicit
rewards_accuracies	0.625	422	2806	explicit
rewards_margins	0.382724	422	2807	explicit
loss	0.915221	423	2809	explicit
rewards_chosen	0.306875	423	2810	explicit
rewards_rejected	-0.0693331	423	2811	explicit
rewards_accuracies	0.5625	423	2812	explicit
rewards_margins	0.376208	423	2813	explicit
loss	0.820414	424	2815	explicit
rewards_chosen	0.363319	424	2816	explicit
rewards_rejected	-0.419662	424	2817	explicit
rewards_accuracies	0.8125	424	2818	explicit
rewards_margins	0.782981	424	2819	explicit
loss	0.907782	425	2821	explicit
rewards_chosen	0.343392	425	2822	explicit
rewards_rejected	-0.0512636	425	2823	explicit
rewards_accuracies	0.75	425	2824	explicit
rewards_margins	0.394656	425	2825	explicit
loss	0.948537	426	2827	explicit
rewards_chosen	0.345052	426	2828	explicit
rewards_rejected	0.126689	426	2829	explicit
rewards_accuracies	0.375	426	2830	explicit
rewards_margins	0.218363	426	2831	explicit
loss	0.879291	427	2833	explicit
rewards_chosen	0.482432	427	2834	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.0252878	427	2835	explicit
rewards_accuracies	0.875	427	2836	explicit
rewards_margins	0.50772	427	2837	explicit
loss	0.883101	428	2839	explicit
rewards_chosen	0.345304	428	2840	explicit
rewards_rejected	-0.172022	428	2841	explicit
rewards_accuracies	0.75	428	2842	explicit
rewards_margins	0.517326	428	2843	explicit
loss	0.970412	429	2845	explicit
rewards_chosen	0.374233	429	2846	explicit
rewards_rejected	0.248569	429	2847	explicit
rewards_accuracies	0.6875	429	2848	explicit
rewards_margins	0.125664	429	2849	explicit
loss	0.876557	430	2851	explicit
rewards_chosen	0.307287	430	2852	explicit
rewards_rejected	-0.207603	430	2853	explicit
rewards_accuracies	0.75	430	2854	explicit
rewards_margins	0.514891	430	2855	explicit
loss	0.920624	431	2857	explicit
rewards_chosen	0.304595	431	2858	explicit
rewards_rejected	-0.0376099	431	2859	explicit
rewards_accuracies	0.6875	431	2860	explicit
rewards_margins	0.342205	431	2861	explicit
loss	0.943717	432	2863	explicit
rewards_chosen	0.108073	432	2864	explicit
rewards_rejected	-0.133759	432	2865	explicit
rewards_accuracies	0.6875	432	2866	explicit
rewards_margins	0.241832	432	2867	explicit
loss	0.947851	433	2869	explicit
rewards_chosen	0.233762	433	2870	explicit
rewards_rejected	0.00411339	433	2871	explicit
rewards_accuracies	0.6875	433	2872	explicit
rewards_margins	0.229649	433	2873	explicit
loss	0.86565	434	2875	explicit
rewards_chosen	0.416582	434	2876	explicit
rewards_rejected	-0.155444	434	2877	explicit
rewards_accuracies	0.75	434	2878	explicit
rewards_margins	0.572027	434	2879	explicit
loss	0.877368	435	2881	explicit
rewards_chosen	0.22408	435	2882	explicit
rewards_rejected	-0.29542	435	2883	explicit
rewards_accuracies	0.75	435	2884	explicit
rewards_margins	0.5195	435	2885	explicit
loss	0.870291	436	2887	explicit
rewards_chosen	0.245053	436	2888	explicit
rewards_rejected	-0.324485	436	2889	explicit
rewards_accuracies	0.6875	436	2890	explicit
rewards_margins	0.569538	436	2891	explicit
loss	0.869907	437	2893	explicit
rewards_chosen	0.31938	437	2894	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.22435	437	2895	explicit
rewards_accuracies	0.9375	437	2896	explicit
rewards_margins	0.54373	437	2897	explicit
loss	0.841492	438	2899	explicit
rewards_chosen	0.352087	438	2900	explicit
rewards_rejected	-0.320676	438	2901	explicit
rewards_accuracies	0.8125	438	2902	explicit
rewards_margins	0.672763	438	2903	explicit
loss	0.955024	439	2905	explicit
rewards_chosen	0.238296	439	2906	explicit
rewards_rejected	0.0568176	439	2907	explicit
rewards_accuracies	0.5625	439	2908	explicit
rewards_margins	0.181478	439	2909	explicit
loss	0.889517	440	2911	explicit
rewards_chosen	0.439331	440	2912	explicit
rewards_rejected	-0.0398899	440	2913	explicit
rewards_accuracies	0.75	440	2914	explicit
rewards_margins	0.479221	440	2915	explicit
loss	0.897407	441	2917	explicit
rewards_chosen	0.206903	441	2918	explicit
rewards_rejected	-0.240673	441	2919	explicit
rewards_accuracies	0.75	441	2920	explicit
rewards_margins	0.447576	441	2921	explicit
loss	0.942924	442	2923	explicit
rewards_chosen	0.179077	442	2924	explicit
rewards_rejected	-0.0663783	442	2925	explicit
rewards_accuracies	0.6875	442	2926	explicit
rewards_margins	0.245455	442	2927	explicit
loss	0.907936	443	2929	explicit
rewards_chosen	0.236874	443	2930	explicit
rewards_rejected	-0.153727	443	2931	explicit
rewards_accuracies	1	443	2932	explicit
rewards_margins	0.3906	443	2933	explicit
loss	0.848387	444	2935	explicit
rewards_chosen	0.468883	444	2936	explicit
rewards_rejected	-0.1718	444	2937	explicit
rewards_accuracies	0.6875	444	2938	explicit
rewards_margins	0.640683	444	2939	explicit
loss	0.867114	445	2941	explicit
rewards_chosen	0.401901	445	2942	explicit
rewards_rejected	-0.173802	445	2943	explicit
rewards_accuracies	0.6875	445	2944	explicit
rewards_margins	0.575703	445	2945	explicit
loss	0.851296	446	2947	explicit
rewards_chosen	0.337915	446	2948	explicit
rewards_rejected	-0.299695	446	2949	explicit
rewards_accuracies	0.6875	446	2950	explicit
rewards_margins	0.63761	446	2951	explicit
loss	0.88122	447	2953	explicit
rewards_chosen	0.323788	447	2954	explicit

Metric	Value	Step	Line	Source
rewards_rejected	-0.197467	447	2955	explicit
rewards_accuracies	0.625	447	2956	explicit
rewards_margins	0.521255	447	2957	explicit
loss	0.893729	448	2959	explicit
rewards_chosen	0.383518	448	2960	explicit
rewards_rejected	-0.058519	448	2961	explicit
rewards_accuracies	0.8125	448	2962	explicit
rewards_margins	0.442037	448	2963	explicit
loss	0.871209	449	2965	explicit
rewards_chosen	0.284048	449	2966	explicit
rewards_rejected	-0.271079	449	2967	explicit
rewards_accuracies	0.75	449	2968	explicit
rewards_margins	0.555127	449	2969	explicit
loss	0.919966	450	2971	explicit
rewards_chosen	0.297516	450	2972	explicit
rewards_rejected	-0.0382172	450	2973	explicit
rewards_accuracies	0.625	450	2974	explicit
rewards_margins	0.335733	450	2975	explicit
eval_rewards_chosen	0.323886	450	2977	explicit
eval_rewards_rejected	-0.0465195	450	2978	explicit
eval_rewards_accuracies	0.680556	450	2979	explicit
eval_rewards_margins	0.370406	450	2980	explicit
eval_loss	0.913172	450	2982	explicit
eval_runtime	77.3167	450	2983	explicit
eval_samples_per_second	4.656	450	2984	explicit
eval_steps_per_second	0.582	450	2985	explicit
eval_entropy	0.887338	450	2986	explicit
eval_num_tokens	2.73854e+07	450	2987	explicit
eval_logits_chosen	10.1351	450	2988	explicit
eval_logits_rejected	10.1624	450	2989	explicit
eval_mean_token_accuracy	0.850733	450	2990	explicit
eval_rewards_chosen	0.323886	450	2991	explicit
eval_rewards_rejected	-0.0465195	450	2992	explicit
eval_rewards_accuracies	0.680556	450	2993	explicit
eval_rewards_margins	0.370406	450	2994	explicit
eval_logps_chosen	-1109.11	450	2995	explicit
eval_logps_rejected	-1111.05	450	2996	explicit
epoch	2.21728	450	2997	explicit
phi	0.847792	450	2998	explicit
phi_bam	0.947662	450	2999	explicit
phi_bmq	0.788389	450	3000	explicit
phi_mku	0.874554	450	3001	explicit
phi_myk	0.858518	450	3002	explicit
phi_ses	0.673515	450	3003	explicit
phi_dtm	0.875799	450	3004	explicit
phi_snk	0.839542	450	3005	explicit
phi_boz	0.880553	450	3006	explicit
phi_ful	0.872129	450	3007	explicit
phi_spp	0.999123	450	3008	explicit
phi_taq	0.715931	450	3009	explicit

Metric	Value	Step	Line	Source
loss	0.853383	451	3010	explicit
rewards_chosen	0.241054	451	3011	explicit
rewards_rejected	-0.38163	451	3012	explicit
rewards_accuracies	0.875	451	3013	explicit
rewards_margins	0.622684	451	3014	explicit
loss	0.950232	452	3016	explicit
rewards_chosen	0.194492	452	3017	explicit
rewards_rejected	-0.0142502	452	3018	explicit
rewards_accuracies	0.625	452	3019	explicit
rewards_margins	0.208742	452	3020	explicit
loss	0.925618	453	3022	explicit
rewards_chosen	0.301101	453	3023	explicit
rewards_rejected	-0.018515	453	3024	explicit
rewards_accuracies	0.875	453	3025	explicit
rewards_margins	0.319616	453	3026	explicit
loss	0.825492	454	3028	explicit
rewards_chosen	0.386221	454	3029	explicit
rewards_rejected	-0.37631	454	3030	explicit
rewards_accuracies	0.75	454	3031	explicit
rewards_margins	0.762531	454	3032	explicit
loss	0.949111	455	3034	explicit
rewards_chosen	0.413759	455	3035	explicit
rewards_rejected	0.185359	455	3036	explicit
rewards_accuracies	0.5	455	3037	explicit
rewards_margins	0.2284	455	3038	explicit
loss	0.913808	456	3040	explicit
rewards_chosen	0.209002	456	3041	explicit
rewards_rejected	-0.158638	456	3042	explicit
rewards_accuracies	0.6875	456	3043	explicit
rewards_margins	0.36764	456	3044	explicit
loss	0.935169	457	3046	explicit
rewards_chosen	0.207854	457	3047	explicit
rewards_rejected	-0.0698723	457	3048	explicit
rewards_accuracies	0.625	457	3049	explicit
rewards_margins	0.277726	457	3050	explicit
loss	0.891633	458	3052	explicit
rewards_chosen	0.598219	458	3053	explicit
rewards_rejected	0.126912	458	3054	explicit
rewards_accuracies	0.75	458	3055	explicit
rewards_margins	0.471307	458	3056	explicit
loss	0.842113	459	3058	explicit
rewards_chosen	0.238093	459	3059	explicit
rewards_rejected	-0.453718	459	3060	explicit
rewards_accuracies	0.8125	459	3061	explicit
rewards_margins	0.69181	459	3062	explicit
loss	0.898263	460	3064	explicit
rewards_chosen	0.300257	460	3065	explicit
rewards_rejected	-0.145537	460	3066	explicit
rewards_accuracies	0.6875	460	3067	explicit
rewards_margins	0.445794	460	3068	explicit

Metric	Value	Step	Line	Source
loss	0.856965	461	3070	explicit
rewards_chosen	0.368927	461	3071	explicit
rewards_rejected	-0.258807	461	3072	explicit
rewards_accuracies	0.75	461	3073	explicit
rewards_margins	0.627734	461	3074	explicit
loss	0.854299	462	3076	explicit
rewards_chosen	0.364056	462	3077	explicit
rewards_rejected	-0.260058	462	3078	explicit
rewards_accuracies	0.8125	462	3079	explicit
rewards_margins	0.624114	462	3080	explicit
loss	0.886512	463	3082	explicit
rewards_chosen	0.375763	463	3083	explicit
rewards_rejected	-0.112567	463	3084	explicit
rewards_accuracies	0.875	463	3085	explicit
rewards_margins	0.48833	463	3086	explicit
loss	0.909088	464	3088	explicit
rewards_chosen	0.301838	464	3089	explicit
rewards_rejected	-0.0803425	464	3090	explicit
rewards_accuracies	0.75	464	3091	explicit
rewards_margins	0.382181	464	3092	explicit
loss	0.952173	465	3094	explicit
rewards_chosen	0.364861	465	3095	explicit
rewards_rejected	0.159541	465	3096	explicit
rewards_accuracies	0.75	465	3097	explicit
rewards_margins	0.205321	465	3098	explicit
loss	0.907227	466	3100	explicit
rewards_chosen	0.0714905	466	3101	explicit
rewards_rejected	-0.3197	466	3102	explicit
rewards_accuracies	1	466	3103	explicit
rewards_margins	0.391191	466	3104	explicit
loss	0.919729	467	3106	explicit
rewards_chosen	0.279875	467	3107	explicit
rewards_rejected	-0.0673492	467	3108	explicit
rewards_accuracies	0.625	467	3109	explicit
rewards_margins	0.347224	467	3110	explicit
loss	0.927331	468	3112	explicit
rewards_chosen	0.291412	468	3113	explicit
rewards_rejected	-0.0247864	468	3114	explicit
rewards_accuracies	0.6875	468	3115	explicit
rewards_margins	0.316198	468	3116	explicit
loss	0.933213	469	3118	explicit
rewards_chosen	0.23991	469	3119	explicit
rewards_rejected	-0.0612574	469	3120	explicit
rewards_accuracies	0.5	469	3121	explicit
rewards_margins	0.301167	469	3122	explicit
loss	0.869211	470	3124	explicit
rewards_chosen	0.421131	470	3125	explicit
rewards_rejected	-0.132975	470	3126	explicit
rewards_accuracies	0.75	470	3127	explicit
rewards_margins	0.554105	470	3128	explicit

Metric	Value	Step	Line	Source
loss	0.801556	471	3130	explicit
rewards_chosen	0.540308	471	3131	explicit
rewards_rejected	-0.329465	471	3132	explicit
rewards_accuracies	0.8125	471	3133	explicit
rewards_margins	0.869772	471	3134	explicit
loss	0.853786	472	3136	explicit
rewards_chosen	0.272289	472	3137	explicit
rewards_rejected	-0.346494	472	3138	explicit
rewards_accuracies	0.8125	472	3139	explicit
rewards_margins	0.618782	472	3140	explicit
loss	0.820898	473	3142	explicit
rewards_chosen	0.512017	473	3143	explicit
rewards_rejected	-0.263323	473	3144	explicit
rewards_accuracies	0.6875	473	3145	explicit
rewards_margins	0.775341	473	3146	explicit
loss	0.93293	474	3148	explicit
rewards_chosen	0.288797	474	3149	explicit
rewards_rejected	0.00420189	474	3150	explicit
rewards_accuracies	0.6875	474	3151	explicit
rewards_margins	0.284595	474	3152	explicit
loss	0.927847	475	3154	explicit
rewards_chosen	0.0807827	475	3155	explicit
rewards_rejected	-0.238328	475	3156	explicit
rewards_accuracies	0.5	475	3157	explicit
rewards_margins	0.319111	475	3158	explicit
loss	0.928028	476	3160	explicit
rewards_chosen	0.354518	476	3161	explicit
rewards_rejected	0.0558397	476	3162	explicit
rewards_accuracies	0.875	476	3163	explicit
rewards_margins	0.298678	476	3164	explicit
loss	0.923792	477	3166	explicit
rewards_chosen	0.264742	477	3167	explicit
rewards_rejected	-0.0584175	477	3168	explicit
rewards_accuracies	0.5625	477	3169	explicit
rewards_margins	0.323159	477	3170	explicit
loss	0.908599	478	3172	explicit
rewards_chosen	0.187948	478	3173	explicit
rewards_rejected	-0.20593	478	3174	explicit
rewards_accuracies	0.6875	478	3175	explicit
rewards_margins	0.393878	478	3176	explicit
loss	0.912784	479	3178	explicit
rewards_chosen	0.222193	479	3179	explicit
rewards_rejected	-0.145591	479	3180	explicit
rewards_accuracies	0.6875	479	3181	explicit
rewards_margins	0.367783	479	3182	explicit
loss	0.953637	480	3184	explicit
rewards_chosen	0.410901	480	3185	explicit
rewards_rejected	0.214384	480	3186	explicit
rewards_accuracies	0.625	480	3187	explicit
rewards_margins	0.196517	480	3188	explicit

Metric	Value	Step	Line	Source
loss	0.96977	481	3190	explicit
rewards_chosen	0.132359	481	3191	explicit
rewards_rejected	0.00972609	481	3192	explicit
rewards_accuracies	0.75	481	3193	explicit
rewards_margins	0.122633	481	3194	explicit
loss	0.935033	482	3196	explicit
rewards_chosen	0.2147	482	3197	explicit
rewards_rejected	-0.0661949	482	3198	explicit
rewards_accuracies	0.875	482	3199	explicit
rewards_margins	0.280894	482	3200	explicit
loss	0.930815	483	3202	explicit
rewards_chosen	0.238863	483	3203	explicit
rewards_rejected	-0.0554636	483	3204	explicit
rewards_accuracies	0.8125	483	3205	explicit
rewards_margins	0.294327	483	3206	explicit
loss	0.898623	484	3208	explicit
rewards_chosen	0.341997	484	3209	explicit
rewards_rejected	-0.0822319	484	3210	explicit
rewards_accuracies	0.75	484	3211	explicit
rewards_margins	0.424229	484	3212	explicit
loss	0.944032	485	3214	explicit
rewards_chosen	0.261837	485	3215	explicit
rewards_rejected	0.0295315	485	3216	explicit
rewards_accuracies	0.75	485	3217	explicit
rewards_margins	0.232306	485	3218	explicit
loss	0.834963	486	3220	explicit
rewards_chosen	0.293712	486	3221	explicit
rewards_rejected	-0.417665	486	3222	explicit
rewards_accuracies	0.9375	486	3223	explicit
rewards_margins	0.711377	486	3224	explicit
loss	0.929372	487	3226	explicit
rewards_chosen	-0.0298473	487	3227	explicit
rewards_rejected	-0.339081	487	3228	explicit
rewards_accuracies	0.5625	487	3229	explicit
rewards_margins	0.309233	487	3230	explicit
loss	0.89616	488	3232	explicit
rewards_chosen	0.26622	488	3233	explicit
rewards_rejected	-0.211014	488	3234	explicit
rewards_accuracies	0.8125	488	3235	explicit
rewards_margins	0.477234	488	3236	explicit
loss	0.896596	489	3238	explicit
rewards_chosen	0.257354	489	3239	explicit
rewards_rejected	-0.186227	489	3240	explicit
rewards_accuracies	0.875	489	3241	explicit
rewards_margins	0.443581	489	3242	explicit
loss	0.938976	490	3244	explicit
rewards_chosen	0.155628	490	3245	explicit
rewards_rejected	-0.108951	490	3246	explicit
rewards_accuracies	0.625	490	3247	explicit
rewards_margins	0.264579	490	3248	explicit

Metric	Value	Step	Line	Source
loss	0.871282	491	3250	explicit
rewards_chosen	0.266816	491	3251	explicit
rewards_rejected	-0.2682	491	3252	explicit
rewards_accuracies	0.75	491	3253	explicit
rewards_margins	0.535016	491	3254	explicit
loss	0.883459	492	3256	explicit
rewards_chosen	0.176442	492	3257	explicit
rewards_rejected	-0.318007	492	3258	explicit
rewards_accuracies	0.625	492	3259	explicit
rewards_margins	0.494449	492	3260	explicit
loss	0.884519	493	3262	explicit
rewards_chosen	0.230702	493	3263	explicit
rewards_rejected	-0.266	493	3264	explicit
rewards_accuracies	0.625	493	3265	explicit
rewards_margins	0.496702	493	3266	explicit
loss	0.9016	494	3268	explicit
rewards_chosen	0.24124	494	3269	explicit
rewards_rejected	-0.176482	494	3270	explicit
rewards_accuracies	0.75	494	3271	explicit
rewards_margins	0.417722	494	3272	explicit
loss	0.847996	495	3274	explicit
rewards_chosen	0.436643	495	3275	explicit
rewards_rejected	-0.210425	495	3276	explicit
rewards_accuracies	0.75	495	3277	explicit
rewards_margins	0.647068	495	3278	explicit
loss	0.932903	496	3280	explicit
rewards_chosen	0.355061	496	3281	explicit
rewards_rejected	0.0662207	496	3282	explicit
rewards_accuracies	0.75	496	3283	explicit
rewards_margins	0.28884	496	3284	explicit
loss	0.936135	497	3286	explicit
rewards_chosen	0.192825	497	3287	explicit
rewards_rejected	-0.0717417	497	3288	explicit
rewards_accuracies	0.8125	497	3289	explicit
rewards_margins	0.264567	497	3290	explicit
loss	0.931706	498	3292	explicit
rewards_chosen	0.150946	498	3293	explicit
rewards_rejected	-0.148543	498	3294	explicit
rewards_accuracies	0.75	498	3295	explicit
rewards_margins	0.299488	498	3296	explicit
loss	0.882046	499	3298	explicit
rewards_chosen	0.0728085	499	3299	explicit
rewards_rejected	-0.429088	499	3300	explicit
rewards_accuracies	0.75	499	3301	explicit
rewards_margins	0.501896	499	3302	explicit
loss	0.899064	500	3304	explicit
rewards_chosen	0.491311	500	3305	explicit
rewards_rejected	0.0567837	500	3306	explicit
rewards_accuracies	0.8125	500	3307	explicit
rewards_margins	0.434527	500	3308	explicit

Metric	Value	Step	Line	Source
eval_rewards_chosen	0.295494	500	3310	explicit
eval_rewards_rejected	-0.0857438	500	3311	explicit
eval_rewards accuracies	0.669444	500	3312	explicit
eval_rewards_margins	0.381238	500	3313	explicit
eval_loss	0.911244	500	3315	explicit
eval_runtime	77.3757	500	3316	explicit
eval_samples_per_second	4.653	500	3317	explicit
eval_steps_per_second	0.582	500	3318	explicit
eval_entropy	0.887046	500	3319	explicit
eval_num_tokens	3.0435e+07	500	3320	explicit
eval_logits_chosen	10.1392	500	3321	explicit
eval_logits_rejected	10.1657	500	3322	explicit
eval_mean_token_accuracy	0.85068	500	3323	explicit
eval_rewards_chosen	0.295494	500	3324	explicit
eval_rewards_rejected	-0.0857438	500	3325	explicit
eval_rewards accuracies	0.669444	500	3326	explicit
eval_rewards_margins	0.381238	500	3327	explicit
eval_logps_chosen	-1109.39	500	3328	explicit
eval_logps_rejected	-1111.44	500	3329	explicit
epoch	2.4642	500	3330	explicit
phi	0.847792	500	3331	explicit
phi_bam	0.947662	500	3332	explicit
phi_bmq	0.788389	500	3333	explicit
phi_mku	0.874554	500	3334	explicit
phi_myk	0.858518	500	3335	explicit
phi_ses	0.673515	500	3336	explicit
phi_dtm	0.875799	500	3337	explicit
phi_snk	0.839542	500	3338	explicit
phi_boz	0.880553	500	3339	explicit
phi_ful	0.872129	500	3340	explicit
phi_spp	0.999123	500	3341	explicit
phi_taq	0.715931	500	3342	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	3	07:29:59	2.4 s	0.73 GB	

Run 4: apo.yaml

Metric	Value	Step	Line	Source
phi_bam	0.823658	0	11	explicit
mer_bam	0	0	12	explicit
mcs_bam	0	0	13	explicit
uwec_bam	1	0	14	explicit
phi_bmq	0.752429	0	15	explicit
mer_bmq	3	0	16	explicit
mcs_bmq	0.833333	0	17	explicit
uwec_bmq	0.666667	0	18	explicit
phi_boz	0.916823	0	19	explicit
mer_boz	0	0	20	explicit
mcs_boz	0	0	21	explicit
uwec_boz	1	0	22	explicit

Metric	Value	Step	Line	Source	
phi_dtm	0.809387	0	23	explicit	
mer_dtm	0.5	0	24	explicit	
mcs_dtm	0.333333	0	25	explicit	
uwec_dtm	0.666667	0	26	explicit	
phi_ful	0.980054	0	27	explicit	
mer_ful	3	0	28	explicit	
mcs_ful	0.75	0	29	explicit	
uwec_ful	0.625	0	30	explicit	
phi_mey	0.768726	0	31	explicit	
mer_mey	0	0	32	explicit	
mcs_mey	0	0	33	explicit	
uwec_mey	0.5	0	34	explicit	
phi_mku	0.908601	0	35	explicit	
mer_mku	12	0	36	explicit	
mcs_mku	0.923077	0	37	explicit	
uwec_mku	0.538462	0	38	explicit	
phi_kao	0.970238	0	39	explicit	
mer_kao	25.3333	0	40	explicit	
mcs_kao	0.974359	0	41	explicit	
uwec_kao	1	0	42	explicit	
phi_myk	0.7953	0	43	explicit	
mer_myk	10	0	44	explicit	
mcs_myk	0.909091	0	45	explicit	
uwec_myk	0.5	0	46	explicit	
phi_ses	0.836952	0	47	explicit	
mer_ses	0	0	48	explicit	
mcs_ses	0	0	49	explicit	
uwec_ses	1	0	50	explicit	
phi_spp	0.853018	0	51	explicit	
mer_spp	6.66667	0	52	explicit	
mcs_spp	0.909091	0	53	explicit	
uwec_spp	0.545455	0	54	explicit	
phi_snk	0.781673	0	55	explicit	
mer_snk	0.333333	0	56	explicit	
mcs_snk	0.333333	0	57	explicit	
uwec_snk	0.666667	0	58	explicit	
phi_taq	0.84844	0	59	explicit	
mer_taq	0.5	0	60	explicit	
mcs_taq	0.5	0	61	explicit	
uwec_taq	1	0	62	explicit	
phi	0.849638	0	63	explicit	
mer	5.16667	0	65	explicit	
mcs	0.865169	0	67	explicit	
uwec	0.769663	0	69	explicit	
Phase	Line	Time	CPU time	Memory	GPU
eval	2	08:33:37	2.3 s	0.68 GB	

B Provenance timeline and activity map

Run 1: kveritas_arm.py

#	Snapshot	Time	Root	Changed
0	run_start	07:29:03	1f46ebf2f29b9b43	136
1	run_end	07:29:04	741ab980f318aa55	1

Event	Path or command	Hash
read	cookbooks/kveritas_arm.py	
read	beni/utils/config.py	
read	beni/utils/res/mdf_fields.json	
read	beni/utils/res/language_metadata.json	
read	beni/utils/res/iso_code_639_3.json	
read	beni/core/__init__.py	
read	beni/core/language.py	
read	beni/core/srl/__init__.py	
read	beni/core/srl/config.py	
read	beni/core/srl/unified.py	
read	beni/core/safety/governor.py	
read	.github/workflows/docs.yml	
read	__init__.py	
read	beni/__init__.py	
read	beni/cli/__init__.py	
read	beni/cli/main.py	
read	beni/core/compute/__init__.py	
read	beni/core/compute/helpers.py	
read	beni/core/compute/metrics.py	
read	beni/core/compute/rewards.py	
read	beni/core/compute/uwec.py	
read	beni/core/hub/__init__.py	
read	beni/core/hub/model_card.py	
read	beni/core/morphotactic/__init__.py	
read	beni/core/morphotactic/dabax.py	
read	beni/core/morphotactic/distil/__init__.py	
read	beni/core/morphotactic/distil/distillation.py	
read	beni/core/morphotactic/distil/providers/__init__.py	
read	beni/core/morphotactic/distil/providers/base.py	
read	beni/core/morphotactic/distil/providers/gguf.py	
read	beni/core/morphotactic/distil/providers/google.py	
read	beni/core/morphotactic/distil/providers/openai_compat.py	
read	beni/core/pipeline.py	
read	beni/core/safety/__init__.py	
read	beni/core/safety/spec.py	
read	beni/core/srl/algorithm1.py	
read	beni/core/srl/apo/__init__.py	
read	beni/core/srl/apo/apo.py	
read	beni/core/srl/dpo/__init__.py	
read	beni/core/srl/dpo/dpo.py	
read	beni/core/srl/grpo/__init__.py	
read	beni/core/srl/grpo/callbacks.py	

Event	Path or command	Hash
read	beni/core/srl/grpo/grpo.py	
read	beni/core/srl/jax/__init__.py	
read	beni/core/srl/jax/policy.py	
read	beni/core/srl/plugin.py	
read	beni/core/srl/sft/__init__.py	
read	beni/core/srl/sft/sft.py	
read	beni/core/wordfreq/__init__.py	
read	beni/data/__init__.py	
read	beni/data/baselines/bmq/ex.json	
read	beni/data/datasets.py	
read	beni/data/exp.yaml	
read	beni/data/test.json	
read	beni/utis/__init__.py	
read	beni/utis/hf_lang.py	
read	beni/utis/res/__init__.py	
read	cleanup.sh	
read	configs/apo_bam.yaml	
read	configs/dpo_bam.yaml	
read	configs/exp.yaml	
read	configs/grpo_bam.yaml	
read	configs/grpo_multilang.yaml	
read	configs/matrix/base_multi13.yaml	
read	configs/matrix/models.json	
read	configs/presets/multi13.yaml	
read	configs/presets/single.yaml	
read	conftest.py	
read	cookbooks/eval_apo.ipynb	
read	cookbooks/eval_dpo.ipynb	
read	cookbooks/eval_grpo.ipynb	
read	cookbooks/eval_sft.ipynb	
read	cookbooks/experimentation_matrix.ipynb	
read	cookbooks/kveritas/dpo.yaml	
read	cookbooks/kveritas/sft.yaml	
read	cookbooks/run_exp.ipynb	
read	docs/javascripts/mathjax.js	
read	docs/javascripts/typewriter.js	
read	experiments/matrix/gemma-3-270m_apo/config.yaml	
read	experiments/matrix/gemma-3-270m_dpo/config.yaml	
read	experiments/matrix/gemma-3-270m_grpo/config.yaml	
read	experiments/matrix/gemma-3-270m_sft/config.yaml	
read	experiments/matrix/gemma-4-e2b_apo/config.yaml	
read	experiments/matrix/gemma-4-e2b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_sft/config.yaml	
read	experiments/matrix/gemma-4-e4b_apo/config.yaml	
read	experiments/matrix/gemma-4-e4b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_sft/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_apo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_dpo/config.yaml	

Event	Path or command	Hash
read	experiments/matrix/granite-4.0-h-350m_grpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_sft/config.yaml	
read	experiments/matrix/lfm2.5-230m_apo/config.yaml	
read	experiments/matrix/lfm2.5-230m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_sft/config.yaml	
read	experiments/matrix/lfm2.5-350m_apo/config.yaml	
read	experiments/matrix/lfm2.5-350m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_sft/config.yaml	
read	experiments/matrix/matrix_results.json	
read	experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_apo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_dpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_grpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_sft/config.yaml	
read	experiments/matrix/qwen3-0.6b_apo/config.yaml	
read	experiments/matrix/qwen3-0.6b_dpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_grpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_sft/config.yaml	
read	experiments/matrix/smollm2-135m_apo/config.yaml	
read	experiments/matrix/smollm2-135m_dpo/config.yaml	
read	experiments/matrix/smollm2-135m_grpo/config.yaml	
read	experiments/matrix/smollm2-135m_sft/config.yaml	
read	experiments/matrix/smollm3-3b_apo/config.yaml	
read	experiments/matrix/smollm3-3b_dpo/config.yaml	
read	experiments/matrix/smollm3-3b_grpo/config.yaml	
read	experiments/matrix/smollm3-3b_sft/config.yaml	
read	experiments/matrix/tinyllama-1.1b_apo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_dpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_grpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_sft/config.yaml	
read	mkdocs.yml	
read	pyproject.toml	
read	scripts/run_matrix.py	
read	setup.cfg	
read	setup.py	
read	tests/__init__.py	
read	tests/test_framework.py	
read	tests/test_grpo_pipeline.py	
read	tests/test_kveritas_callback.py	
write	cookbooks/kveritas/apo.yaml	3906c6b5c6b5f5ec
process	/tmp/kv-sebeni/venv/bin/python3.12 /tmp/kv-sebeni/venv/bin/sebeni init --lan...	

Run 2: apo.yaml

#	Snapshot	Time	Root	Changed
0	run_start	07:29:05	741ab980f318aa55	137
1	phase distill	07:29:07	741ab980f318aa55	0
2	run_end	07:29:57	741ab980f318aa55	0

Event	Path or command	Hash
read	beni/cli/__init__.py	
read	beni/cli/main.py	
read	beni/core/language.py	
read	beni/utils/res/mdf_fields.json	
read	beni/utils/res/language_metadata.json	
read	beni/utils/res/iso_code_639_3.json	
read	beni/core/srl/__init__.py	
read	beni/core/srl/config.py	
read	beni/core/srl/unified.py	
read	.github/workflows/docs.yml	
read	__init__.py	
read	beni/__init__.py	
read	beni/core/__init__.py	
read	beni/core/compute/__init__.py	
read	beni/core/compute/rewards.py	
read	beni/core/hub/__init__.py	
read	beni/core/hub/model_card.py	
read	beni/core/morphotactic/__init__.py	
read	beni/core/morphotactic/distil/distillation.py	
read	beni/core/morphotactic/distil/providers/gguf.py	
read	beni/core/morphotactic/distil/providers/google.py	
read	beni/core/morphotactic/distil/providers/openai_compat.py	
read	beni/core/pipeline.py	
read	beni/core/safety/__init__.py	
read	beni/core/safety/governor.py	
read	beni/core/srl/algorithm1.py	
read	beni/core/srl/apo/__init__.py	
read	beni/core/srl/apo/apo.py	
read	beni/core/srl/dpo/__init__.py	
read	beni/core/srl/dpo/dpo.py	
read	beni/core/srl/grpo/__init__.py	
read	beni/core/srl/grpo/callbacks.py	
read	beni/core/srl/grpo/grpo.py	
read	beni/core/srl/jax/__init__.py	
read	beni/core/srl/jax/policy.py	
read	beni/core/srl/plugin.py	
read	beni/core/srl/sft/__init__.py	
read	beni/core/srl/sft/sft.py	
read	beni/core/wordfreq/__init__.py	
read	beni/data/baselines/bmq/ex.json	
read	beni/data/exp.yaml	
read	beni/data/test.json	
read	beni/utils/__init__.py	
read	beni/utils/config.py	

Event	Path or command	Hash
read	beni/utills/res/__init__.py	
read	cleanup.sh	
read	configs/apo_bam.yaml	
read	configs/dpo_bam.yaml	
read	configs/exp.yaml	
read	configs/grpo_bam.yaml	
read	configs/grpo_multilang.yaml	
read	configs/matrix/base_multi13.yaml	
read	configs/matrix/models.json	
read	configs/presets/multi13.yaml	
read	configs/presets/single.yaml	
read	conftest.py	
read	cookbooks/eval_apo.ipynb	
read	cookbooks/eval_dpo.ipynb	
read	cookbooks/eval_grpo.ipynb	
read	cookbooks/eval_sft.ipynb	
read	cookbooks/experimentation_matrix.ipynb	
read	cookbooks/kveritas/apo.yaml	
read	cookbooks/kveritas/dpo.yaml	
read	cookbooks/kveritas/sft.yaml	
read	cookbooks/kveritas_arm.py	
read	cookbooks/run_exp.ipynb	
read	docs/javascripts/mathjax.js	
read	docs/javascripts/typewriter.js	
read	experiments/matrix/gemma-3-270m_apo/config.yaml	
read	experiments/matrix/gemma-3-270m_dpo/config.yaml	
read	experiments/matrix/gemma-3-270m_grpo/config.yaml	
read	experiments/matrix/gemma-3-270m_sft/config.yaml	
read	experiments/matrix/gemma-4-e2b_apo/config.yaml	
read	experiments/matrix/gemma-4-e2b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_sft/config.yaml	
read	experiments/matrix/gemma-4-e4b_apo/config.yaml	
read	experiments/matrix/gemma-4-e4b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_sft/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_apo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_dpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_grpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_sft/config.yaml	
read	experiments/matrix/lfm2.5-230m_apo/config.yaml	
read	experiments/matrix/lfm2.5-230m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_sft/config.yaml	
read	experiments/matrix/lfm2.5-350m_apo/config.yaml	
read	experiments/matrix/lfm2.5-350m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_sft/config.yaml	
read	experiments/matrix/matrix_results.json	
read	experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml	

Event	Path or command	Hash
read	experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_apo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_dpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_grpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_sft/config.yaml	
read	experiments/matrix/qwen3-0.6b_apo/config.yaml	
read	experiments/matrix/qwen3-0.6b_dpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_grpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_sft/config.yaml	
read	experiments/matrix/smollm2-135m_apo/config.yaml	
read	experiments/matrix/smollm2-135m_dpo/config.yaml	
read	experiments/matrix/smollm2-135m_grpo/config.yaml	
read	experiments/matrix/smollm2-135m_sft/config.yaml	
read	experiments/matrix/smollm3-3b_apo/config.yaml	
read	experiments/matrix/smollm3-3b_dpo/config.yaml	
read	experiments/matrix/smollm3-3b_grpo/config.yaml	
read	experiments/matrix/smollm3-3b_sft/config.yaml	
read	experiments/matrix/tinyllama-1.1b_apo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_dpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_grpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_sft/config.yaml	
read	mkdocs.yml	
read	pyproject.toml	
read	scripts/run_matrix.py	
read	setup.cfg	
read	setup.py	
read	tests/__init__.py	
read	tests/test_framework.py	
read	tests/test_grpo_pipeline.py	
read	tests/test_kveritas_callback.py	
read	beni/core/safety/spec.py	
read	beni/data/__init__.py	
read	beni/data/datasets.py	
read	beni/data/raw/dataset_300_samples.jsonl	
read	beni/core/morphotactic/distil/__init__.py	
read	beni/utis/hf_lang.py	
read	beni/core/morphotactic/distil/providers/__init__.py	
read	beni/core/morphotactic/distil/providers/base.py	
read	beni/core/compute/metrics.py	
read	beni/core/compute/helpers.py	
read	beni/core/compute/uwec.py	
read	beni/data/baselines/spp/baseline.gram	
read	beni/data/baselines/spp/baseline.dict	
read	beni/core/morphotactic/dabax.py	
read	beni/data/baselines/bam/baseline.gram	
read	beni/data/baselines/bam/baseline.dict	
read	beni/data/baselines/taq/baseline.gram	
read	beni/data/baselines/taq/baseline.dict	

Event	Path or command	Hash
read	beni/data/baselines/bmq/baseline.gram	
read	beni/data/baselines/bmq/baseline.dict	
read	beni/data/baselines/mku/baseline.gram	
read	beni/data/baselines/mku/baseline.dict	
read	beni/data/baselines/dtm/baseline.gram	
read	beni/data/baselines/dtm/baseline.dict	
read	beni/data/baselines/myk/baseline.gram	
read	beni/data/baselines/myk/baseline.dict	
read	beni/data/baselines/boz/baseline.gram	
read	beni/data/baselines/boz/baseline.dict	
read	beni/data/baselines/ses/baseline.gram	
read	beni/data/baselines/ses/baseline.dict	
read	beni/data/baselines/ful/baseline.gram	
read	beni/data/baselines/ful/baseline.dict	
read	beni/data/baselines/snk/baseline.gram	
read	beni/data/baselines/snk/baseline.dict	

Run 3: 0.1

#	Snapshot	Time	Root	Changed
0	run_start	07:29:58	741ab980f318aa55	137
1	phase train	07:29:59	741ab980f318aa55	0
2	run_end	08:33:35	741ab980f318aa55	0

Event	Path or command	Hash
read	beni/__init__.py	
read	beni/cli/__init__.py	
read	beni/core/__init__.py	
read	beni/core/language.py	
read	beni/utils/config.py	
read	beni/utils/res/mdf_fields.json	
read	beni/utils/res/language_metadata.json	
read	beni/utils/res/iso_code_639_3.json	
read	beni/core/srl/config.py	
read	beni/core/srl/unified.py	
read	.github/workflows/docs.yml	
read	__init__.py	
read	beni/cli/main.py	
read	beni/core/compute/__init__.py	
read	beni/core/compute/metrics.py	
read	beni/core/compute/uwec.py	
read	beni/core/morphotactic/dabax.py	
read	beni/core/morphotactic/distil/providers/base.py	
read	beni/core/morphotactic/distil/providers/gguf.py	
read	beni/core/morphotactic/distil/providers/google.py	
read	beni/core/morphotactic/distil/providers/openai_compat.py	
read	beni/core/pipeline.py	
read	beni/core/safety/__init__.py	
read	beni/core/safety/governor.py	
read	beni/core/safety/spec.py	
read	beni/core/srl/__init__.py	

Event	Path or command	Hash
read	beni/core/srl/algorithm1.py	
read	beni/core/srl/apo/__init__.py	
read	beni/core/srl/dpo/__init__.py	
read	beni/core/srl/jax/__init__.py	
read	beni/core/srl/jax/policy.py	
read	beni/core/srl/plugin.py	
read	beni/core/srl/sft/sft.py	
read	beni/core/wordfreq/__init__.py	
read	beni/data/__init__.py	
read	beni/data/baselines/bmq/ex.json	
read	beni/data/datasets.py	
read	beni/data/exp.yaml	
read	beni/data/test.json	
read	beni/utis/__init__.py	
read	beni/utis/hf_lang.py	
read	beni/utis/res/__init__.py	
read	cleanup.sh	
read	configs/apo_bam.yaml	
read	configs/dpo_bam.yaml	
read	configs/exp.yaml	
read	configs/grpo_bam.yaml	
read	configs/grpo_multilang.yaml	
read	configs/matrix/base_multi13.yaml	
read	configs/matrix/models.json	
read	configs/presets/multi13.yaml	
read	configs/presets/single.yaml	
read	conftest.py	
read	cookbooks/eval_apo.ipynb	
read	cookbooks/eval_dpo.ipynb	
read	cookbooks/eval_grpo.ipynb	
read	cookbooks/eval_sft.ipynb	
read	cookbooks/experimentation_matrix.ipynb	
read	cookbooks/kveritas/apo.yaml	
read	cookbooks/kveritas/dpo.yaml	
read	cookbooks/kveritas/sft.yaml	
read	cookbooks/kveritas_arm.py	
read	cookbooks/run_exp.ipynb	
read	docs/javascripts/mathjax.js	
read	docs/javascripts/typewriter.js	
read	experiments/matrix/gemma-3-270m_apo/config.yaml	
read	experiments/matrix/gemma-3-270m_dpo/config.yaml	
read	experiments/matrix/gemma-3-270m_grpo/config.yaml	
read	experiments/matrix/gemma-3-270m_sft/config.yaml	
read	experiments/matrix/gemma-4-e2b_apo/config.yaml	
read	experiments/matrix/gemma-4-e2b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_sft/config.yaml	
read	experiments/matrix/gemma-4-e4b_apo/config.yaml	
read	experiments/matrix/gemma-4-e4b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_grpo/config.yaml	

Event	Path or command	Hash
read	experiments/matrix/gemma-4-e4b_sft/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_apo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_dpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_grpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_sft/config.yaml	
read	experiments/matrix/lfm2.5-230m_apo/config.yaml	
read	experiments/matrix/lfm2.5-230m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_sft/config.yaml	
read	experiments/matrix/lfm2.5-350m_apo/config.yaml	
read	experiments/matrix/lfm2.5-350m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_sft/config.yaml	
read	experiments/matrix/matrix_results.json	
read	experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_apo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_dpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_grpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_sft/config.yaml	
read	experiments/matrix/qwen3-0.6b_apo/config.yaml	
read	experiments/matrix/qwen3-0.6b_dpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_grpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_sft/config.yaml	
read	experiments/matrix/smollm2-135m_apo/config.yaml	
read	experiments/matrix/smollm2-135m_dpo/config.yaml	
read	experiments/matrix/smollm2-135m_grpo/config.yaml	
read	experiments/matrix/smollm2-135m_sft/config.yaml	
read	experiments/matrix/smollm3-3b_apo/config.yaml	
read	experiments/matrix/smollm3-3b_dpo/config.yaml	
read	experiments/matrix/smollm3-3b_grpo/config.yaml	
read	experiments/matrix/smollm3-3b_sft/config.yaml	
read	experiments/matrix/tinyllama-1.1b_apo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_dpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_grpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_sft/config.yaml	
read	mkdocs.yml	
read	pyproject.toml	
read	scripts/run_matrix.py	
read	setup.cfg	
read	setup.py	
read	tests/__init__.py	
read	tests/test_framework.py	
read	tests/test_grpo_pipeline.py	
read	tests/test_kveritas_callback.py	
read	beni/data/raw/dataset_300_samples.jsonl	
read	beni/core/srl/grpo/__init__.py	
read	beni/core/srl/grpo/callbacks.py	

Event	Path or command	Hash
read	beni/core/srl/grpo/grpo.py	
read	beni/core/compute/rewards.py	
read	beni/core/compute/helpers.py	
read	beni/core/morphotactic/__init__.py	
read	beni/core/hub/__init__.py	
read	beni/core/hub/model_card.py	
read	beni/core/morphotactic/distil/__init__.py	
read	beni/core/morphotactic/distil/distillation.py	
read	beni/core/morphotactic/distil/providers/__init__.py	
read	beni/core/srl/dpo/dpo.py	
read	beni/core/srl/apo/apo.py	
read	beni/core/srl/sft/__init__.py	
read	beni/core/hub/templates/model_card.md.j2	

Run 4: apo.yaml

#	Snapshot	Time	Root	Changed
0	run_start	08:33:36	741ab980f318aa55	137
1	phase eval	08:33:37	741ab980f318aa55	0
2	run_end	08:35:39	741ab980f318aa55	0

Event	Path or command	Hash
read	beni/cli/__init__.py	
read	beni/core/language.py	
read	beni/utls/__init__.py	
read	beni/utls/config.py	
read	beni/utls/res/mdf_fields.json	
read	beni/utls/res/language_metadata.json	
read	beni/utls/res/iso_code_639_3.json	
read	beni/core/srl/__init__.py	
read	beni/core/srl/unified.py	
read	beni/core/safety/governor.py	
read	.github/workflows/docs.yml	
read	__init__.py	
read	beni/__init__.py	
read	beni/cli/main.py	
read	beni/core/__init__.py	
read	beni/core/compute/__init__.py	
read	beni/core/compute/metrics.py	
read	beni/core/compute/uwec.py	
read	beni/core/hub/__init__.py	
read	beni/core/hub/model_card.py	
read	beni/core/morphotactic/__init__.py	
read	beni/core/morphotactic/distil/providers/gguf.py	
read	beni/core/morphotactic/distil/providers/google.py	
read	beni/core/morphotactic/distil/providers/openai_compat.py	
read	beni/core/safety/__init__.py	
read	beni/core/srl/algorithm1.py	
read	beni/core/srl/apo/__init__.py	
read	beni/core/srl/apo/apo.py	
read	beni/core/srl/config.py	

Event	Path or command	Hash
read	beni/core/srl/dpo/dpo.py	
read	beni/core/srl/grpo/__init__.py	
read	beni/core/srl/grpo/callbacks.py	
read	beni/core/srl/jax/__init__.py	
read	beni/core/srl/jax/policy.py	
read	beni/core/srl/sft/__init__.py	
read	beni/core/srl/sft/sft.py	
read	beni/core/wordfreq/__init__.py	
read	beni/data/__init__.py	
read	beni/data/baselines/bmq/ex.json	
read	beni/data/datasets.py	
read	beni/data/exp.yaml	
read	beni/data/test.json	
read	beni/utlis/hf_lang.py	
read	beni/utlis/res/__init__.py	
read	cleanup.sh	
read	configs/apo_bam.yaml	
read	configs/dpo_bam.yaml	
read	configs/exp.yaml	
read	configs/grpo_bam.yaml	
read	configs/grpo_multilang.yaml	
read	configs/matrix/base_multi13.yaml	
read	configs/matrix/models.json	
read	configs/presets/multi13.yaml	
read	configs/presets/single.yaml	
read	conftest.py	
read	cookbooks/eval_apo.ipynb	
read	cookbooks/eval_dpo.ipynb	
read	cookbooks/eval_grpo.ipynb	
read	cookbooks/eval_sft.ipynb	
read	cookbooks/experimentation_matrix.ipynb	
read	cookbooks/kveritas/apo.yaml	
read	cookbooks/kveritas/dpo.yaml	
read	cookbooks/kveritas/sft.yaml	
read	cookbooks/kveritas_arm.py	
read	cookbooks/run_exp.ipynb	
read	docs/javascripts/mathjax.js	
read	docs/javascripts/typewriter.js	
read	experiments/matrix/gemma-3-270m_apo/config.yaml	
read	experiments/matrix/gemma-3-270m_dpo/config.yaml	
read	experiments/matrix/gemma-3-270m_grpo/config.yaml	
read	experiments/matrix/gemma-3-270m_sft/config.yaml	
read	experiments/matrix/gemma-4-e2b_apo/config.yaml	
read	experiments/matrix/gemma-4-e2b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e2b_sft/config.yaml	
read	experiments/matrix/gemma-4-e4b_apo/config.yaml	
read	experiments/matrix/gemma-4-e4b_dpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_grpo/config.yaml	
read	experiments/matrix/gemma-4-e4b_sft/config.yaml	

Event	Path or command	Hash
read	experiments/matrix/granite-4.0-h-350m_apo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_dpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_grpo/config.yaml	
read	experiments/matrix/granite-4.0-h-350m_sft/config.yaml	
read	experiments/matrix/lfm2.5-230m_apo/config.yaml	
read	experiments/matrix/lfm2.5-230m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-230m_sft/config.yaml	
read	experiments/matrix/lfm2.5-350m_apo/config.yaml	
read	experiments/matrix/lfm2.5-350m_dpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_grpo/config.yaml	
read	experiments/matrix/lfm2.5-350m_sft/config.yaml	
read	experiments/matrix/matrix_results.json	
read	experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml	
read	experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_apo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_dpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_grpo/config.yaml	
read	experiments/matrix/qwen2.5-0.5b_sft/config.yaml	
read	experiments/matrix/qwen3-0.6b_apo/config.yaml	
read	experiments/matrix/qwen3-0.6b_dpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_grpo/config.yaml	
read	experiments/matrix/qwen3-0.6b_sft/config.yaml	
read	experiments/matrix/smollm2-135m_apo/config.yaml	
read	experiments/matrix/smollm2-135m_dpo/config.yaml	
read	experiments/matrix/smollm2-135m_grpo/config.yaml	
read	experiments/matrix/smollm2-135m_sft/config.yaml	
read	experiments/matrix/smollm3-3b_apo/config.yaml	
read	experiments/matrix/smollm3-3b_dpo/config.yaml	
read	experiments/matrix/smollm3-3b_grpo/config.yaml	
read	experiments/matrix/smollm3-3b_sft/config.yaml	
read	experiments/matrix/tinyllama-1.1b_apo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_dpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_grpo/config.yaml	
read	experiments/matrix/tinyllama-1.1b_sft/config.yaml	
read	mkdocs.yml	
read	pyproject.toml	
read	scripts/run_matrix.py	
read	setup.cfg	
read	setup.py	
read	tests/__init__.py	
read	tests/test_framework.py	
read	tests/test_grpo_pipeline.py	
read	tests/test_kveritas_callback.py	
read	beni/core/safety/spec.py	
read	beni/core/pipeline.py	
read	beni/core/srl/grpo/grpo.py	
read	beni/core/srl/plugin.py	

Event	Path or command	Hash
read	beni/core/compute/rewards.py	
read	beni/core/compute/helpers.py	
read	beni/core/morphotactic/dabax.py	
read	beni/core/morphotactic/distil/__init__.py	
read	beni/core/morphotactic/distil/distillation.py	
read	beni/core/morphotactic/distil/providers/__init__.py	
read	beni/core/morphotactic/distil/providers/base.py	
read	beni/core/srl/dpo/__init__.py	
read	beni/data/baselines/mey/baseline.gram	
read	beni/data/baselines/mey/baseline.dict	
read	beni/data/baselines/kao/baseline.gram	
read	beni/data/baselines/kao/baseline.dict	

C Run history

Invocation 1, OK, 1.8 seconds, 6 output lines, started 2026-10-05T07:29:03

```
metric_hash e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
run_digest e258decd1924b981171b674ef9c0218b9d23b0b741f53fe7c986a61ef8ace3a5
anchored   at run end
```

Invocation 2, OK, 51.8 seconds, 36 output lines, started 2026-10-05T07:29:05

```
metric_hash e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
run_digest fbeda81e470356edaddcb7d438c5d1daaef75d852fd03f62c56583b614055713
anchored   at run end
```

Invocation 3, OK, 1 hours 3 minutes, 3344 output lines, started 2026-10-05T07:29:57

```
metric_hash 106d946950e38efd87212da2aab2b7e3823d0029f4958b516d1689b2801dffa2
run_digest 0fca4b909a90f76e2cc82f2553afdd5babe199321446235823cf0b82c429c6e9
anchored   at run end
```

Invocation 4, OK, 2 minutes 3 seconds, 70 output lines, started 2026-10-05T08:33:35

```
metric_hash a30c845a667bb06b397e7909692da3f173e9108a9897806a21aad0715bf595c2
run_digest 88f80b01ee1c3f36fbae631d54e9792622a0a8e85d4fcf6614a16723193ff257
anchored   at run end
```

D Raw seal fields and hashes

Seal

```
data_hash      3d35b38ed6fb3a13c0ac03d020b070894bb4f1d5469f322ef456aca5b4eff9a6
signed_message_hash 4daeeaa62541a57716bc022707750b515cc2f56dce152d5fe3c010dbaaa6ef3f9
nonce          37bd3b1eff5d9dbb727af8210253c7a6
signed_at      2026-10-05T08:35:40.609502+00:00
sealed_at      2026-10-05T08:35:40.67932167Z
checkout_bundle 62f040330bb576322043e72745134b995ff5b696bdfa1f8c62fa006027af6947
```

Signature (base64)

```
s2jq7G40TF3dm5bTxwXfUfW82GlbJDHzu00wq3BG+NmIM/SgzOd+dlH90rRoq45qKjx8rc/il0K6X1iUEe56k
vBYmoLuE4Xg3VlMH9uqVsXERZy9mSkdKD6ZjIEKADh1JDv+N0q4wMbus7xefvx2wr5vRostQwYQ7VKNvT0YoFeVz
euIPRQSkCX2if2CJ6kE910tEuPxFBK/qHcYCvi6yqc8GwoFiK4JXULBm+XVfVNBmnKsvVwZlwVPay5jvZmx/ZdM2
WUoJPVwkG0nGzKowUnSRuJfY1cVT00AEH3RTVFTbu7oZiNGDs7+wgpp8KZT/BCJSD10aIrm1BVt08+8xYGNXwvk7
0Szn0hyHf6lno9s4FVodCcMID4vqHSLBzkd2o2A4su/v2gqSKmjquXMwTZe8k83iee24YfBKU1U+stWkt8CXV0nn
zEwjphnocoHVol9wnovv90A43bYTrkiZoZJJKkuq8OMC4+Xf9TfkmGeX14E0cdbZeb4PFPTCBDSMfeozVbH2RBXP
3CM4n40jz8bYsmBo0Lnz4+MGFc9o/9Ao7zs9foMNVa29UKssKEiN0/8x6R0LdEikWMx8B4e40QojgyuZah5XlFeD
z1/86merZIRLiB0idBH+72/oCieKdP956QFcwIvTUadruBqq73EQLX0y9Fma8gKkQR0=
```

Public key

```

-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAt4vcgWP+iWscPc3HzpmjpsKbTmOemqgp6RB6GnnPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEKjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMGicNgeNHNCy9bLgpU5annzP9gN2l
6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/oLCHBCf1VIDCkDxohehdZ0WRRCgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----

```

Run 1 outputs

```

stdout  9565359eac9ba2d2259f970e74339f9ebab4ecc47a34c82fa0b9158ca6b7a208
stderr  e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
digest  e258dedc1924b981171b674ef9c0218b9d23b0b741f53fe7c986a61ef8ace3a5
cookbooks/kveritas_arm.py (unchanged)
pre     c221e36799c1e2d1e6ea95a6fd46e514b4f99444da247695e531964b3b67c67d
post    c221e36799c1e2d1e6ea95a6fd46e514b4f99444da247695e531964b3b67c67d

```

Run 2 outputs

```

stdout  38e526ccff7eb23bf14c1dc7cf806b55a4149ec538afa74332ce3cfe55b9f3bd
stderr  e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
digest  fbada81e470356edaddcb7d438c5d1daaef75d852fd03f62c56583b614055713

```

Run 3 outputs

```

stdout  81cca997fc5e604a6864259f519926cf736f6ef2c26ad5b624eff6cd42bfa61e
stderr  9851c251e32841468d20a7e0eb92e42a5b1549a7940e88bcfc56ff283ad347b8
metrics 106d946950e38efd87212da2aab2b7e3823d0029f4958b516d1689b2801dffa2
digest  0fca4b909a90f76e2cc82f2553afdd5babe199321446235823cf0b82c429c6e9

```

Run 4 outputs

```

stdout  a0b2a7330f34ac632473bd777700d6025e09fea15238f3ace784e69d67ded8a0
stderr  3a3dbbaa93e186edc81e26185ff21a3389fd6227fa35ef3dd9ec4ea87f9a85b9
metrics a30c845a667bb06b397e7909692da3f173e9108a9897806a21aad0715bf595c2
digest  88f80b01ee1c3f36fbae631d54e9792622a0a8e85d4fcf6614a16723193ff257

```

E BibTeX for [1]

```

@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}

```